Complex Analysis: Contours, Fourier Transforms, and Schwarz–Christoffel

Problems with background and complete solutions

Principal-branch integral of Log^2 along a curve from i to $-i$

Problem. Let $\gamma : [0,1] \to \mathbb{C}$ be a C^1 curve with $\gamma(0) = i$, $\gamma(1) = -i$, such that γ does not intersect $(-\infty, 0]$. Compute

$$\int_\gamma \operatorname{Log}^2(z)\, dz,$$

where Log denotes the principal branch.

Background. We use $\operatorname{Log} z = \ln|z| + i \operatorname{Arg} z$ with branch cut $(-\infty, 0]$ and $\operatorname{Arg} z \in (-\pi, \pi)$. On any simply connected set avoiding the cut, $\operatorname{Log}^2 z$ is holomorphic, hence the integral is path independent. Seek an antiderivative of the form $F(z) = z\, G(\operatorname{Log} z)$. Then

$$F'(z) = G(\operatorname{Log} z) + G'(\operatorname{Log} z) = \operatorname{Log}^2 z \iff G'(w) + G(w) = w^2.$$

A polynomial solution is $G(w) = w^2 - 2w + 2$, since $(2w-2) + (w^2 - 2w + 2) = w^2$. Thus

$$F(z) = z\big(\operatorname{Log}^2 z - 2\operatorname{Log} z + 2\big), \qquad F'(z) = \operatorname{Log}^2 z.$$

Evaluation. Using $\operatorname{Log}(i) = i\frac{\pi}{2}$, $\operatorname{Log}(-i) = -i\frac{\pi}{2}$,

$$F(i) = i\left(\left(i\frac{\pi}{2}\right)^2 - 2\left(i\frac{\pi}{2}\right) + 2\right) = -\frac{i\pi^2}{4} + \pi + 2i,$$

$$F(-i) = -i\left(\left(-i\frac{\pi}{2}\right)^2 - 2\left(-i\frac{\pi}{2}\right) + 2\right) = \frac{i\pi^2}{4} + \pi - 2i.$$

Therefore

$$\boxed{\int_\gamma \operatorname{Log}^2 z\, dz = F(-i) - F(i) = i\left(\frac{\pi^2}{2} - 4\right).}$$

Remark. More generally, for $n \in \mathbb{N}$ one may integrate $\operatorname{Log}^n z$ by solving $G' + G = w^n$ and taking $F(z) = z\, G(\operatorname{Log} z)$.

Integral of $e^{e^{it}}$ over one period

Problem. Compute $\displaystyle\int_0^{2\pi} e^{e^{it}}\, dt$.

Path method. Let $z = e^{it}$ so that $|z| = 1$ and $dt = \frac{dz}{iz}$. Then

$$\int_0^{2\pi} e^{e^{it}}\, dt = \frac{1}{i} \oint_{|z|=1} \frac{e^z}{z}\, dz.$$

By Cauchy's integral formula for $f(z) = e^z$,

$$\oint_{|z|=1} \frac{e^z}{z}\, dz = 2\pi i\, f(0) = 2\pi i.$$

Hence

$$\boxed{\int_0^{2\pi} e^{\,e^{it}}\, dt = 2\pi}.$$

Remark (series check). Since $e^{e^{it}} = \sum_{n=0}^{\infty} \frac{e^{int}}{n!}$, the integral over $[0, 2\pi]$ kills all terms with $n \neq 0$ and equals 2π.

Cauchy's Theorem from Green's Theorem

Setup. Let C be a positively oriented simple closed C^1 curve bounding $D \subset \mathbb{R}^2$. Assume $f = u + iv$ is holomorphic on an open set containing $\overline{D}$ and that u_x, u_y, v_x, v_y are continuous on D.

Tools. Green's theorem: $\int_C (L\, dx + M\, dy) = \iint_D (M_x - L_y)\, dx\, dy$. For holomorphic $f = u + iv$, the Cauchy–Riemann equations hold: $u_x = v_y$, $u_y = -v_x$. Also $dz = dx + i\, dy$, hence

$$\int_C f\, dz = \int_C (u\, dx - v\, dy) + i \int_C (v\, dx + u\, dy).$$

Proof. Apply Green to each real 1–form.

Real part: with $L = u$, $M = -v$,

$$\int_C (u\, dx - v\, dy) = \iint_D \big[(-v)_x - u_y\big]\, dx\, dy = \iint_D (-v_x - u_y)\, dx\, dy = 0$$

since $u_y = -v_x$.

Imaginary part: with $L = v$, $M = u$,

$$\int_C (v\, dx + u\, dy) = \iint_D (u_x - v_y)\, dx\, dy = 0$$

since $u_x = v_y$.

Therefore $\int_C f(z)\, dz = 0$.

Bonus (why not this proof?). This argument requires C^1 regularity of u, v and invokes Green's theorem from vector calculus. In complex analysis we prefer Goursat's approach, which proves Cauchy's theorem without assuming continuity of partial derivatives (only complex differentiability), and it applies to piecewise C^1 curves and more general domains. It is thus strictly more general and self-contained within complex analysis.

Cauchy's Theorem from Green's Theorem — extended background

Holomorphic integrand as a 1–form. For $z = x + iy$ and $f = u + iv$,

$$f\,dz = (u\,dx - v\,dy) + i\,(v\,dx + u\,dy),$$

so $\displaystyle\int_C f\,dz$ splits into two real line integrals.

Green as Stokes in 2D. Green's theorem is the special case of Stokes for 1–forms:

$$\oint_{\partial D} (L\,dx + M\,dy) = \iint_D (M_x - L_y)\,dx \wedge dy = \iint_D d(L\,dx + M\,dy).$$

Thus a closed 1–form (one with vanishing exterior derivative) integrates to 0 on any closed curve.

CR $\Rightarrow$ closedness. If f is holomorphic, the Cauchy–Riemann equations hold: $u_x = v_y$ and $u_y = -v_x$. Then

$$d(u\,dx - v\,dy) = (-v_x - u_y)\,dx \wedge dy = 0, \qquad d(v\,dx + u\,dy) = (u_x - v_y)\,dx \wedge dy = 0.$$

Applying Green to each form gives

$$\int_C (u\,dx - v\,dy) = 0, \qquad \int_C (v\,dx + u\,dy) = 0,$$

hence $\displaystyle\int_C f\,dz = 0$.

Complex-coordinate proof (optional). With $dz = dx + i\,dy$, $d\bar{z} = dx - i\,dy$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$, one has

$$d(f\,dz) = (\partial_{\bar{z}} f)\,d\bar{z} \wedge dz.$$

If f is holomorphic, $\partial_{\bar{z}} f = 0$, so by Stokes

$$\oint_{\partial D} f\,dz = \iint_D d(f\,dz) = 0.$$

Why not use this as the main proof? This argument assumes $u, v \in C^1$ and invokes Green's theorem. Goursat's theorem proves Cauchy's theorem without any continuity of partials, for piecewise C^1 curves, and stays within complex methods, which is both more general and conceptually central to the theory.

Green's Theorem: deeper background and the "calc" to Cauchy

Green/Stokes in several forms. Let $D \subset \mathbb{R}^2$ have positively oriented, piecewise C^1 boundary ∂D and let $L, M \in C^1(\overline{D})$. Then

$$\oint_{\partial D} (L\,dx + M\,dy) = \iint_D \big(M_x - L_y\big)\,dx\,dy.$$

Equivalently, for the 1–form $\alpha = L\,dx + M\,dy$,

$$\oint_{\partial D} \alpha = \iint_D d\alpha, \qquad d\alpha = (M_x - L_y)\,dx \wedge dy.$$

If D has holes, orient the outer boundary counterclockwise and each inner boundary clockwise; the sum of the boundary integrals equals the same area integral.

Regularity. We assume $L, M \in C^1$ and piecewise C^1 boundary; in our application this will follow from u_x, u_y, v_x, v_y being continuous.

Complex 1–forms and $\partial_{\bar z}$. With $z = x + iy$, $dz = dx + i\,dy$, $d\bar z = dx - i\,dy$, $\partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y)$,

$$d(f\,dz) = (\partial_{\bar z} f)\,d\bar z \wedge dz.$$

Thus $d(f\,dz) = 0$ iff $\partial_{\bar z} f = 0$, i.e. f is holomorphic (CR).

Cauchy from Green: the calculation. Let $f = u + iv$ be holomorphic with u_x, u_y, v_x, v_y continuous on $\overline{D}$. Since

$$\int_{\partial D} f\,dz = \int_{\partial D} (u\,dx - v\,dy) + i \int_{\partial D} (v\,dx + u\,dy),$$

apply Green twice:

Real part: with $L = u$, $M = -v$,

$$\int_{\partial D} (u\,dx - v\,dy) = \iint_D (-v_x - u_y)\,dx\,dy = 0 \quad (\text{since } u_y = -v_x).$$

Imaginary part: with $L = v$, $M = u$,

$$\int_{\partial D} (v\,dx + u\,dy) = \iint_D (u_x - v_y)\,dx\,dy = 0 \quad (\text{since } u_x = v_y).$$

Therefore

$$\boxed{\int_{\partial D} f(z)\,dz = 0.}$$

Remarks. On multiply connected regions, the sum of boundary integrals (with orientations) vanishes. Goursat's theorem proves the same result without assuming the continuity of the first partials, which is why it is preferred as the foundational proof in complex analysis.

Practical 1-forms and calculations (plane and beyond)

A. Plane examples

A1. Area as a line integral. For $\alpha = x\,dy$ or $\tilde\alpha = \frac{1}{2}(x\,dy - y\,dx)$,

$$d\alpha = d\tilde\alpha = dx \wedge dy, \qquad \oint_{\partial D} x\,dy = \iint_D 1\,dx\,dy = \text{Area}(D), \quad \oint_{\partial D} \tfrac{1}{2}(x\,dy - y\,dx) = \text{Area}(D).$$

Ellipse: For $C:\ x^2/a^2 + y^2/b^2 = 1$, $\oint_C \frac{1}{2}(x\,dy - y\,dx) = \pi ab$.

A2. Polynomial field via Green. Let $\omega = (x^2 - y)\,dx + (x + y^2)\,dy$. Then

$$d\omega = (x + y^2)_x - (x^2 - y)_y = 2.$$

On the unit disk, $\displaystyle\oint_{x^2+y^2=1} \omega = \iint_{x^2+y^2\le 1} 2\,dx\,dy = 2\pi.$

A3. Closed vs exact. A 1-form $\omega = P\,dx + Q\,dy$ is closed if $P_y = Q_x$. On simply connected domains, closed $\Rightarrow$ exact.

Exact: $\omega = 2x\,dx + 2y\,dy = d(x^2 + y^2)$, hence $\oint \omega = 0$.

Closed but not exact (punctured plane):

$$\eta = \frac{-y}{x^2 + y^2}\,dx + \frac{x}{x^2 + y^2}\,dy, \qquad d\eta = 0 \text{ on } \mathbb{R}^2 \backslash \{0\}.$$

Yet $\displaystyle\oint_{x^2+y^2=1} \eta = 2\pi.$

B. Complex-coordinate viewpoint

B1. dz, $d\bar{z}$ and $\partial_{\bar{z}}$. With $z = x + iy$, $dz = dx + i\,dy$, $d\bar{z} = dx - i\,dy$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$,

$$d(f\,dz) = (\partial_{\bar{z}} f)\,d\bar{z} \wedge dz.$$

If f is holomorphic, then $d(f\,dz) = 0$ and $\oint_{\partial D} f\,dz = 0$.

B2. Singularities (glimpse of residues). For $f(z) = \frac{1}{z-a}$ and C a small CCW circle about a,

$$\oint_C f(z)\,dz = 2\pi i.$$

B3. Multiply connected domains. If D is an annulus with CCW outer boundary C_{out} and clockwise inner boundary C_{in}, then for holomorphic f on D,

$$\oint_{C_{\text{out}}} f\,dz + \oint_{C_{\text{in}}} f\,dz = 0.$$

C. Polygon area (shoelace via Green).

For vertices (x_k, y_k) in CCW order and $(x_{n+1}, y_{n+1}) = (x_1, y_1)$,

$$\text{Area}(D) = \oint_{\partial D} x\,dy = \frac{1}{2}\sum_{k=1}^{n} \big(x_k y_{k+1} - x_{k+1} y_k\big).$$

Tips. To evaluate $\oint_C (P\,dx + Q\,dy)$: check $M_x - L_y$. If simple, use Green; if 0, try a potential or inspect topology (holes). In complex settings, convert to $\oint \frac{f(z)}{z-a}\,dz$ when possible for Cauchy/Residue; otherwise Cauchy's theorem gives 0.

A) Warm-ups on the plane

A1) Area from a line integral

Take the 1-form

$$\alpha = x\,dy,$$

or the symmetrized area form

$$\tilde{\alpha} = \tfrac{1}{2}\big(x\,dy - y\,dx\big).$$

Then

$$d\alpha = dx \wedge dy,$$

and

$$d\tilde{\alpha} = dx \wedge dy.$$

By Green's theorem,

$$\oint_{\partial D} x\,dy \;=\; \iint_D 1\,dx\,dy,$$

hence

$$\oint_{\partial D} x\,dy \;=\; \mathrm{Area}(D).$$

Also,

$$\oint_{\partial D} \tfrac{1}{2}\big(x\,dy - y\,dx\big) \;=\; \iint_D dx \wedge dy,$$

so

$$\oint_{\partial D} \tfrac{1}{2}\big(x\,dy - y\,dx\big) \;=\; \mathrm{Area}(D).$$

Example (ellipse). For the ellipse

$$C: \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

we have

$$\oint_C \tfrac{1}{2}\big(x\,dy - y\,dx\big) \;=\; \pi ab.$$

(You can parametrize $x = a\cos t$, $y = b\sin t$ and compute directly, or just invoke Green's theorem.)

A2) Circulation and flux of a polynomial vector field

Let

$$\omega \;=\; (x^2 - y)\,dx \;+\; (x + y^2)\,dy.$$

Then

$$d\omega \;=\; \big(\partial_x(x + y^2) - \partial_y(x^2 - y)\big)\,dx \wedge dy,$$

so

$$d\omega \;=\; \big(1 - (-1)\big)\,dx \wedge dy \;=\; 2\,dx \wedge dy.$$

Over the unit circle $C : x^2 + y^2 = 1$ (and unit disk),

$$\oint_C \omega \;=\; \iint_{x^2+y^2\le 1} 2\,dx\,dy,$$

hence

$$\oint_C \omega \;=\; 2\,\pi.$$

A3) Exact vs. closed: quick tests

For a 1-form $\omega = P\,dx + Q\,dy$,

$$\omega \text{ is closed} \iff P_y = Q_x.$$

If D is simply connected, then

$$\text{closed} \;\Rightarrow\; \text{exact}, \quad \text{i.e. } \exists\,\phi \text{ with } \omega = d\phi.$$

Exact example.

$$\omega \;=\; 2x\,dx + 2y\,dy \;=\; d(x^2 + y^2).$$

Therefore any loop integral satisfies

$$\oint \omega \;=\; 0,$$

either by Green (since $d\omega = 0$) or by the fundamental theorem for line integrals.

Closed but not exact on the punctured plane. Define

$$\eta \;=\; \frac{-y}{x^2+y^2}\,dx \;+\; \frac{x}{x^2+y^2}\,dy.$$

Then

$$d\eta \;=\; 0 \quad \text{on } \mathbb{R}^2 \setminus \{0\},$$

but on the unit circle $|z| = 1$,

$$\oint_{|z|=1} \eta \;=\; 2\pi.$$

This nonzero loop integral reflects the topological obstruction (the "hole" at the origin). It is the real-variable analogue of

$$\oint \frac{1}{z}\,dz \;=\; 2\pi i.$$

B) Complex-coordinate computations

B1) Using dz, $d\bar{z}$ and $\partial_{\bar{z}}$

For smooth f,

$$d\big(f\,dz\big) \;=\; \big(\partial_{\bar{z}} f\big)\,d\bar{z} \wedge dz.$$

If f is holomorphic, that is $\partial_{\bar{z}} f = 0$, then the 1-form $f\,dz$ is closed, and by Stokes/Green we have

$$\oint_{\partial D} f\,dz \;=\; 0,$$

whenever f is holomorphic on a region containing D.

Example. Let $f(z) = z^2$ and C any closed C^1 loop avoiding singularities (there are none). Then

$$\oint_C z^2\,dz \;=\; 0.$$

B2) Holomorphic with an isolated singularity (residues peek)

Let

$$f(z) \;=\; \frac{1}{z-a},$$

and let C be a positively oriented circle centered at a. Then

$$\oint_C f(z)\,dz \;=\; 2\pi i.$$

In real-form language, the real part of $\frac{1}{z}\,dz$ is exactly the η from A3:

$$\Re\!\left(\frac{1}{z}\,dz\right) \;=\; \eta,$$

while

$$\Im\!\left(\frac{1}{z}\,dz\right) \;=\; -\,\frac{x}{x^2+y^2}\,dx \;-\; \frac{y}{x^2+y^2}\,dy.$$

Together these give the complex integral $2\pi i$.

B3) Multiply connected domain and orientations

Let D be an annulus with outer boundary C_{out} oriented counterclockwise and inner boundary C_{in} oriented clockwise. If f is holomorphic on D, then

$$\oint_{C_{\text{out}}} f\,dz \;+\; \oint_{C_{\text{in}}} f\,dz \;=\; 0.$$

For example, for $f(z) = z$ each integral is zero. For $f(z) = 1/z$ the above identity does not apply (there is a singularity inside), and each circle integral equals $\pm 2\pi i$ according to the orientation.

C) Quick polygon area (shoelace via Green)

Given a simple polygon with vertices (x_k, y_k) in counterclockwise order (for $k = 1, \ldots, n$), and setting $(x_{n+1}, y_{n+1}) = (x_1, y_1)$, the area can be written as

$$\text{Area}(D) \;=\; \oint_{\partial D} x\,dy,$$

and also as the shoelace sum

$$\mathrm{Area}(D) \;=\; \frac{1}{2}\sum_{k=1}^{n}\Bigl(x_k\,y_{k+1} \;-\; x_{k+1}\,y_k\Bigr).$$

This is just Green's theorem with the choice $\alpha = x\,dy$.

Further tips

To decide a line integral

$$\oint_C \bigl(P\,dx + Q\,dy\bigr),$$

first compute

$$M_x - L_y \;=\; \partial_x Q \;-\; \partial_y P.$$

If $M_x - L_y$ is constant or simple, use Green's theorem immediately. If $M_x - L_y = 0$, try to find a potential ϕ (exactness) or check the topology of the domain for holes that can obstruct exactness.

In complex problems, it is often useful to rewrite integrals in the Cauchy/Residue style

$$\frac{1}{i}\oint \frac{f(z)}{z-a}\,dz,$$

and apply Cauchy's integral formula or the residue theorem when singularities are present. If there are no singularities in the region bounded by the curve, then by Cauchy's theorem

$$\oint f(z)\,dz \;=\; 0.$$

Goal

Explain why

$$\mathrm{Area}(D) \;=\; \frac{1}{2}\oint_{\partial D}\bigl(x\,dy - y\,dx\bigr),$$

including three independent derivations: (1) averaging two Green identities, (2) an exterior-calculus choice of a 1-form with $d\alpha = dx \wedge dy$, (3) a parametrization check leading to the shoelace formula.

Throughout, $D \subset \mathbb{R}^2$ is a region with positively oriented (counterclockwise) boundary ∂D.

1) Green's theorem by "averaging"

Green's theorem states that for smooth $L(x, y), M(x, y)$,

$$\oint_{\partial D}\bigl(L\,dx + M\,dy\bigr) \;=\; \iint_D \bigl(M_x - L_y\bigr)\,dx\,dy.$$

First choice $(L, M) = (0, x)$

Then

$$M_x - L_y \;=\; \partial_x(x) - \partial_y(0) \;=\; 1,$$

so

$$\oint_{\partial D} x\,dy \;=\; \iint_D 1\,dx\,dy \;=\; \mathrm{Area}(D).$$

Second choice $(L, M) = (-y, 0)$

Then

$$M_x - L_y \;=\; \partial_x(0) - \partial_y(-y) \;=\; 1,$$

hence

$$-\oint_{\partial D} y\,dx \;=\; \iint_D 1\,dx\,dy \;=\; \mathrm{Area}(D).$$

Average the two identities

Add the last two equalities and divide by 2:

$$\mathrm{Area}(D) \;=\; \frac{1}{2}\oint_{\partial D} x\,dy \;+\; \frac{1}{2}\Big(-\oint_{\partial D} y\,dx\Big) \;=\; \frac{1}{2}\oint_{\partial D}\big(x\,dy - y\,dx\big).$$

Interpretation. The factor $\frac{1}{2}$ is the average of two *already correct* area-from-boundary formulas: $\displaystyle\oint x\,dy$ and $\displaystyle -\oint y\,dx$.

2) Exterior calculus: pick α with $d\alpha = dx \wedge dy$

We seek a 1-form α of the form

$$\alpha \;=\; a\,x\,dy \;+\; b\,(-y\,dx),$$

with constants $a, b \in \mathbb{R}$, such that $d\alpha$ equals the area 2-form $dx \wedge dy$.

Compute $d\alpha$

Using $d(f\,\beta) = df \wedge \beta + f\,d\beta$, $d(dx) = d(dy) = 0$, and $dy \wedge dx = -dx \wedge dy$:

$$d\alpha \;=\; a\,dx \wedge dy \;+\; b\big(-dy \wedge dx\big) \;=\; (a+b)\,dx \wedge dy.$$

Match the area 2-form

We require

$$d\alpha \;=\; dx \wedge dy, \qquad \Longleftrightarrow \qquad a + b \;=\; 1.$$

A particularly symmetric choice is $a = b = \frac{1}{2}$, yielding

$$\alpha \;=\; \frac{1}{2}\,(x\,dy - y\,dx), \qquad d\alpha \;=\; dx \wedge dy.$$

Apply Stokes/Green

By Stokes,

$$\oint_{\partial D} \alpha \;=\; \iint_D d\alpha \;=\; \iint_D dx \wedge dy \;=\; \mathrm{Area}(D),$$

i.e.

$$\mathrm{Area}(D) \;=\; \frac{1}{2} \oint_{\partial D} \big(x\,dy - y\,dx\big).$$

Interpretation. The factor $\frac{1}{2}$ comes from the symmetric coefficients $a = b = \frac{1}{2}$ that make $d\alpha = dx \wedge dy$.

3) Parametrization check and the shoelace formula

Let the boundary $C = \partial D$ be parametrized (CCW) by $t \mapsto (x(t), y(t))$, $t \in [a, b]$. Then

$$dx \;=\; x'(t)\,dt, \qquad dy \;=\; y'(t)\,dt.$$

Therefore,

$$\oint_{\partial D} \big(x\,dy - y\,dx\big) \;=\; \int_a^b \big(x(t)\,y'(t) - y(t)\,x'(t)\big)\,dt.$$

By Green (or a divergence argument), $\frac{1}{2}$ of this integral equals the area:

$$\mathrm{Area}(D) \;=\; \frac{1}{2} \int_a^b \big(x\,y' - y\,x'\big)\,dt \;=\; \frac{1}{2} \oint_{\partial D} \big(x\,dy - y\,dx\big).$$

Polygonal boundary: the shoelace

If C is a simple polygon with vertices (x_k, y_k) in CCW order for $k = 1, \ldots, n$, and $(x_{n+1}, y_{n+1}) = (x_1, y_1)$, then

$$\mathrm{Area}(D) \;=\; \frac{1}{2} \sum_{k=1}^{n} \Big(x_k\,y_{k+1} \;-\; x_{k+1}\,y_k\Big),$$

which is exactly the discrete version of $\frac{1}{2} \oint (x\,dy - y\,dx)$ (the "shoelace formula").

Geometric intuition for the factor $\frac{1}{2}$. Along each small edge element, the signed quantity $x\,dy - y\,dx$ is the oriented area of the infinitesimal parallelogram spanned by (x, y) and (dx, dy). Each such parallelogram is twice the area of the corresponding "triangle strip" between the edge and the origin, hence the factor $\frac{1}{2}$.

Summary (which form to use?)

For a CCW-oriented boundary,

$$\boxed{\mathrm{Area}(D) \;=\; \oint_{\partial D} x\,dy \;=\; -\oint_{\partial D} y\,dx \;=\; \frac{1}{2} \oint_{\partial D} \big(x\,dy - y\,dx\big).}$$

The symmetric form $\frac{1}{2} \oint (x\,dy - y\,dx)$ is rotation-invariant in look and often numerically more stable (errors in the two terms tend to cancel).

Conventions

We write a 1-form on the plane as $\omega = L\,dx + M\,dy$. For a positively oriented (CCW) boundary $C = \partial D$, Green's theorem says

$$\oint_C (L\,dx + M\,dy) \;=\; \iint_D \big(\partial_x M - \partial_y L\big)\,dx\,dy.$$

For complex line integrals on a circle, we often parametrize $z = Re^{it}$, $dz = iRe^{it}\,dt$.

1) Straight segment: direct parametrization

Compute $\displaystyle\int_C (y\,dx + x\,dy)$ for the segment from $(0,0)$ to $(1,2)$.

Parametrization. Let $r(t) = (x(t), y(t)) = (t, 2t)$, $t \in [0,1]$. Then $dx = dt$ and $dy = 2\,dt$.

Integrand.

$$y\,dx + x\,dy = (2t)\,dt + (t)\,(2\,dt) = 4t\,dt.$$

Integral.

$$\int_C (y\,dx + x\,dy) = \int_0^1 4t\,dt = 2.$$

Remark. Since $y\,dx + x\,dy = d(xy)$, the value equals xy at the endpoint minus at the start: $xy\big|_{(0,0)}^{(1,2)} = 2 - 0 = 2$ (path-independent).

2) "Angle" 1-form around a circle (gives 2π)

Let

$$\eta = \frac{-y\,dx + x\,dy}{x^2+y^2}, \qquad C:\; x = \cos t,\; y = \sin t,\; t \in [0, 2\pi].$$

Differentials and denominator. $dx = -\sin t\,dt$, $dy = \cos t\,dt$, and $x^2 + y^2 = 1$.

Integrand.

$$\eta = \frac{-y\,dx + x\,dy}{1} = (-\sin t)(-\sin t)\,dt + (\cos t)(\cos t)\,dt = \big(\sin^2 t + \cos^2 t\big)\,dt = dt.$$

Integral.

$$\oint_C \eta = \int_0^{2\pi} dt = 2\pi.$$

(Real twin of the complex identity $\displaystyle\oint \frac{dz}{z} = 2\pi i$.)

3) Use Green's theorem instead of piecewise parametrization

Let $\omega = (x^2 - y)\,dx + (x + y^2)\,dy$ on the triangle with vertices $(0,0) \to (1,0) \to (1,1) \to (0,0)$.

Curl (scalar).

$d\omega = (\partial_x M - \partial_y L)\,dx \wedge dy = \big(\partial_x(x+y^2) - \partial_y(x^2 - y)\big)\,dx \wedge dy = (1 - (-1))\,dx \wedge dy = 2\,dx \wedge dy.$

Area of D. Right triangle of legs 1: $\text{Area}(D) = \frac{1}{2}$.

Integral.

$$\oint_{\partial D} \omega = \iint_D 2\,dx\,dy = 2 \cdot \frac{1}{2} = 1.$$

4) Conservative field (potential method)

$$\omega = 2x\,dx + 3y^2\,dy = d(x^2 + y^3) = d\phi, \qquad \phi(x,y) = x^2 + y^3.$$

Any path from $A = (0,1)$ to $B = (2,2)$:

$$\int_A^B \omega = \phi(B) - \phi(A) = \big(2^2 + 2^3\big) - \big(0^2 + 1^3\big) = (4+8) - (0+1) = 11.$$

5) Area via $\displaystyle\frac{1}{2}\oint (x\,dy - y\,dx)$ (ellipse)

Ellipse parametrization: $x = 3\cos t$, $y = 2\sin t$, $t \in [0, 2\pi]$. Then $dx = -3\sin t\,dt$, $dy = 2\cos t\,dt$.

Integrand.

$$x\,dy - y\,dx = (3\cos t)(2\cos t)\,dt - (2\sin t)(-3\sin t)\,dt = 6\big(\cos^2 t + \sin^2 t\big)\,dt = 6\,dt.$$

Area.

$$\text{Area} = \frac{1}{2}\int_0^{2\pi} 6\,dt = 6\pi,$$

which matches $\pi ab = \pi \cdot 3 \cdot 2 = 6\pi$.

6) Complex line integrals on a circle

Let $C_R : \ |z| = R$, with $z = Re^{it}$, $dz = iRe^{it}\,dt$.

Case $n \neq -1$.

$$\oint_{C_R} z^n\,dz = \int_0^{2\pi} (Re^{it})^n\, iRe^{it}\,dt = iR^{n+1}\int_0^{2\pi} e^{i(n+1)t}\,dt = 0.$$

Case $n = -1$.

$$\oint_{C_R} \frac{dz}{z} = \int_0^{2\pi} i\,dt = 2\pi i.$$

7) Curved path (parabola) — direct calculation

Compute $\displaystyle\int_C \big(x^2y\,dx + x\,y\,dy\big)$ along $y = x^2$ for $x : 0 \to 1$.

Parametrization. $x = t$, $y = t^2$, so $dx = dt$, $dy = 2t\,dt$.

Pieces.

$$x^2y\,dx = (t^2)(t^2)\,dt = t^4\,dt, \qquad x\,y\,dy = t\cdot t^2\cdot(2t\,dt) = 2t^4\,dt.$$

Integral.

$$\int_C \big(x^2y\,dx + x\,y\,dy\big) = \int_0^1 (t^4 + 2t^4)\,dt = \int_0^1 3t^4\,dt = \frac{3}{5}.$$

8) Circulation of $F = (-y, x)$ around the unit square

Let $\omega = -y\,dx + x\,dy$. Then

$$d\omega = (\partial_x x - \partial_y(-y))\,dx\wedge dy = (1-(-1))\,dx\wedge dy = 2\,dx\wedge dy.$$

Domain. Unit square has area 1.

Integral.

$$\oint_{\partial D} \omega = \iint_D 2\,dx\,dy = 2.$$

Quick recipe (what to try first)

- If the field is given as $\omega = L\,dx + M\,dy$, compute $M_x - L_y$.
  - If $M_x - L_y$ is constant or simple, jump to Green's theorem.
  - If $M_x - L_y = 0$ on a simply connected domain, try a potential ϕ with $\omega = d\phi$ (exactness).
  - If there is a hole (punctures), expect nonzero loop integrals even when $M_x - L_y = 0$.
- For complex integrals on circles, parametrize $z = Re^{it}$ or use Cauchy/Residue when singularities lie inside; otherwise use Cauchy's theorem ($\oint f(z)\,dz = 0$).

Why is it a line segment? We parametrize

$$x(t) = t, \qquad y(t) = 2t, \qquad t \in [0,1].$$

Then $y = 2x$ along the curve and the endpoints are $(0,0)$ at $t = 0$ and $(1,2)$ at $t = 1$. Hence $\{(t,2t) : 0 \le t \le 1\}$ is precisely the straight line segment on the line $y = 2x$ between $(0,0)$ and $(1,2)$.

What does the integral compute? Consider the 1-form

$$\omega \;=\; y\,dx + x\,dy.$$

Since

$$d(xy) \;=\; x\,dy + y\,dx \;=\; \omega,$$

the form ω is exact with potential $\phi(x,y) = xy$. Therefore for any path C from A to B,

$$\int_C \omega \;=\; \phi(B) - \phi(A) \;=\; (xy)\Big|_A^B.$$

For the segment from $(0,0)$ to $(1,2)$,

$$\int_C (y\,dx + x\,dy) \;=\; (xy)\Big|_{(0,0)}^{(1,2)} \;=\; (1\cdot 2) - (0\cdot 0) \;=\; 2.$$

Because ω is exact, the value 2 is the same for any path joining these endpoints.

Line segment C, the 1-form, and why the integral is "work"

What is C? C is the path of integration. Here it is the straight line segment from $(0,0)$ to $(1,2)$. A convenient parametrization is

$$x(t) = t, \qquad y(t) = 2t, \qquad t \in [0,1],$$

so that the image is exactly the portion of the line $y = 2x$ joining the two endpoints. Orientation: increasing t runs from $(0,0)$ to $(1,2)$.

What is the 1-form? A (differential) 1-form on the plane has the shape

$$\omega \;=\; P(x,y)\,dx \;+\; Q(x,y)\,dy.$$

Here

$$\omega \;=\; y\,dx \;+\; x\,dy.$$

Why does it represent "work"? A vector field $\mathbf{F} = (P, Q)$ does work along a path C given by the line integral

$$\int_C \mathbf{F}\cdot d\mathbf{r} \;=\; \int_C \big(P\,dx + Q\,dy\big).$$

Thus integrating the 1-form $P\,dx + Q\,dy$ is the same as computing the mechanical work of $\mathbf{F} = (P, Q)$. In our case $\mathbf{F} = (y, x)$, so

$$\int_C (y\,dx + x\,dy) \;=\; \int_C \mathbf{F}\cdot d\mathbf{r}.$$

Key simplification (exactness). Note that

$$d(xy) \;=\; x\,dy + y\,dx \;=\; \omega,$$

so ω is *exact* with potential $\phi(x, y) = xy$ (i.e. $\omega = d\phi$). By the fundamental theorem for line integrals,

$$\int_C \omega \;=\; \phi(\text{end}) - \phi(\text{start}),$$

which depends only on the endpoints, not on the particular path.

Evaluation for this C. With start $A = (0, 0)$ and end $B = (1, 2)$,

$$\int_C (y\,dx + x\,dy) \;=\; \phi(B) - \phi(A) \;=\; (xy)\Big|_{(0,0)}^{(1,2)} \;=\; (1\cdot 2) - (0\cdot 0) \;=\; 2.$$

Therefore, the value 2 would be the same for *any* path from $(0, 0)$ to $(1, 2)$, not just the straight segment.

Problem

For $\gamma(a, r)$ the positively oriented circle centered at a of radius $r > 0$, compute

$$\int_{\gamma(0,1)} \frac{5z^2 - 8}{z^3 - 2z^2}\,dz.$$

Then describe closed paths γ for which $\displaystyle\int_\gamma f(z)\,dz$ equals **(a)** $14\pi i$ and **(b)** $-2\pi i$, where

$$f(z) = \frac{5z^2 - 8}{z^3 - 2z^2}.$$

Background / Plan

Factor the denominator:

$$z^3 - 2z^2 = z^2(z-2),$$

so f has an order-2 pole at $z = 0$ and a simple pole at $z = 2$.

Use partial fractions to read off residues. For any closed path γ,

$$\int_\gamma f(z)\,dz \;=\; 2\pi i \sum_{a\in\{0,2\}} n(\gamma,a)\ \mathrm{Res}(f;a),$$

where $n(\gamma, a)$ is the winding number of γ about a.

Partial Fractions and Residues

Seek constants A, B, C with

$$\frac{5z^2-8}{z^2(z-2)} \;=\; \frac{A}{z-2} + \frac{B}{z} + \frac{C}{z^2}.$$

Clearing denominators gives

$$5z^2 - 8 = Az^2 + Bz(z-2) + C(z-2) = (A+B)z^2 + (-2B+C)z - 2C.$$

Match coefficients:

$$A + B = 5, \qquad -2B + C = 0, \qquad -2C = -8.$$

Hence $C = 4$, $B = 2$, $A = 3$, so

$$f(z) = \frac{3}{z-2} + \frac{2}{z} + \frac{4}{z^2}.$$

Therefore

$$\mathrm{Res}(f;0) = 2, \qquad \mathrm{Res}(f;2) = 3.$$

(The $4/z^2$ term contributes no residue at 0.)

Integral over $\gamma(0, 1)$

The unit circle about the origin encloses $z = 0$ but not $z = 2$. Thus

$$\int_{\gamma(0,1)} f(z)\,dz = 2\pi i \cdot \mathrm{Res}(f;0) = 2\pi i \cdot 2 = \boxed{4\pi i}.$$

Paths giving prescribed values

For a general closed curve γ with winding numbers

$$n_0 := n(\gamma,0), \qquad n_2 := n(\gamma,2),$$

the residue theorem gives

$$\int_\gamma f(z)\,dz = 2\pi i\,(2n_0 + 3n_2).$$

(a) $14\pi i$. Require $2n_0 + 3n_2 = 7$. Examples (infinitely many):

$$n_0 = 2,\ n_2 = 1 \quad \text{(two CCW loops around 0 and one CCW loop around 2)},$$

$$n_0 = 5,\ n_2 = -1 \quad \text{(five CCW around 0, one CW around 2)}.$$

(b) $-2\pi i$. Require $2n_0 + 3n_2 = -1$. Examples:

$$n_0 = 1,\ n_2 = -1 \quad \text{(one CCW around 0, one CW around 2)},$$

$$n_0 = 4,\ n_2 = -3 \quad \text{(four CCW around 0, three CW around 2)}.$$

Any integer pair (n_0, n_2) solving the linear Diophantine equation yields a valid γ (e.g. a concatenation of small circles around 0 and 2 with the prescribed orientations).

What Green's Theorem Actually Says

Let $D \subset \mathbb{R}^2$ be a region with positively oriented (counterclockwise) boundary $C = \partial D$, and let L, M be C^1 functions on an open set containing $\overline{D}$. Green's theorem states

$$\oint_C \big(L\,dx + M\,dy\big) \;=\; \iint_D \big(M_x - L_y\big)\,dx\,dy.$$

Interpretation: the curve in the line integral must be *the boundary of the region* appearing in the double integral.

1) $D =$ unit disk, $C =$ its boundary (unit circle). Green applies directly:

$$\oint_{|z|=1} \big(L\,dx + M\,dy\big) \;=\; \iint_{\{|z|\le 1\}} \big(M_x - L_y\big)\,dx\,dy.$$

2) $D =$ unit disk, but C is a larger circle (radius $R > 1$). Here C is *not* the boundary of D. The boundary of the unit disk is the *unit* circle, not the larger one. To compare the two line integrals, use the multiply connected version on the annulus

$$A = \{\, 1 \le |z| \le R \,\}.$$

Let C_1 be the unit circle oriented CCW and C_R the circle of radius R oriented CCW. Then Stokes/Green on A gives

$$\oint_{C_R} \big(L\,dx + M\,dy\big) \;-\; \oint_{C_1} \big(L\,dx + M\,dy\big) \;=\; \iint_A \big(M_x - L_y\big)\,dx\,dy.$$

Conclusion: the two line integrals are equal iff the area integral over the annulus is zero— e.g. if $M_x - L_y \equiv 0$ on A (the 1-form is closed there). Otherwise, they differ by

$$\iint_A \big(M_x - L_y\big)\,dx\,dy,$$

the integral of the curl over the region between the circles.

3) Complex-analysis corollary. Let $f = u + iv$ be holomorphic on a region containing the annulus between two closed curves (with no singularities inside). The Cauchy–Riemann equations imply the 1-forms

$$u\,dx - v\,dy, \qquad v\,dx + u\,dy$$

are *closed*, i.e. their curls vanish:

$$\partial_x(v) - \partial_y(u) = 0, \qquad \partial_x(u) + \partial_y(v) = 0.$$

Hence their line integrals are the same on any two homologous curves (e.g. the unit circle and a larger circle) provided the region between them contains no singularities of f. This is the path-independence you use when applying Cauchy's theorem.

Green in 2D: what's what?

Region D. The (closed) disk of radius R centered at the origin:

$$D \;=\; \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \le R^2\}.$$

Boundary $C = \partial D$. The positively oriented circle:

$$C: \quad x = R\cos t, \quad y = R\sin t, \qquad t \in [0, 2\pi].$$

Vector field / function. Let $\mathbf{F} = (L, M)$ with $L = L(x,y)$, $M = M(x,y)$.

Associated 1-form.

$$\alpha \;=\; L\,dx + M\,dy.$$

Green's theorem.

$$\underbrace{\oint_C \big(L\,dx + M\,dy\big)}_{\text{line integral (circulation)}} \;=\; \underbrace{\iint_D \big(M_x - L_y\big)\,dx\,dy}_{\text{scalar curl (“rotation”)}}.$$

What each side measures

$$\oint_C \big(L\,dx + M\,dy\big) \;=\; \int_C \mathbf{F}\cdot\mathbf{T}\,ds \quad \text{(total tangential circulation/work along } C\text{)},$$

$$\iint_D \big(M_x - L_y\big)\,dA \quad \text{(total vorticity / scalar curl over the interior)}.$$

Green says: circulation around the edge equals total curl inside.

Example 1 — constant curl, easy numbers

Take $\mathbf{F} = (-y, x)$. Then $L = -y$, $M = x$ and

$$M_x - L_y \;=\; \partial_x(x) - \partial_y(-y) \;=\; 1 - (-1) \;=\; 2 \quad \text{(constant)}.$$

Area integral:

$$\iint_D (M_x - L_y)\, dA \;=\; \iint_D 2\, dA \;=\; 2\,\pi R^2.$$

Line integral (check by parametrizing C):

$$dx = -R\sin t\, dt, \qquad dy = R\cos t\, dt,$$

$$L\, dx + M\, dy = (-y)\, dx + (x)\, dy = (-R\sin t)(-R\sin t)\, dt + (R\cos t)(R\cos t)\, dt = R^2\, dt,$$

$$\oint_C (L\, dx + M\, dy) = \int_0^{2\pi} R^2\, dt = 2\pi R^2.$$

Both sides match: $2\pi R^2$.

Example 2 — gradient field (curl $= 0$), circulation $= 0$

Let $\phi(x, y) = x^2 + y^3$ and $\mathbf{F} = \nabla\phi = (2x,\, 3y^2)$. Then

$$M_x - L_y = \partial_x(3y^2) - \partial_y(2x) = 0 - 0 = 0,$$

so

$$\oint_C (2x\, dx + 3y^2\, dy) = \iint_D 0\, dA = 0.$$

Interpretation: exact/gradient fields do no net work around closed curves.

Example 3 — area from a 1-form (what $\frac{1}{2}(x\,dy - y\,dx)$ measures)

Set

$$\alpha \;=\; \frac{1}{2}\,(x\, dy - y\, dx).$$

Then

$$d\alpha = \frac{1}{2}\,(dx \wedge dy + dy \wedge dx) = dx \wedge dy.$$

Hence, by Stokes/Green,

$$\oint_C \alpha = \iint_D dx \wedge dy = \text{Area}(D) = \pi R^2.$$

Therefore,

$$\boxed{\frac{1}{2}\oint_C (x\, dy - y\, dx) \;=\; \text{Area}(D) \;=\; \pi R^2}.$$

Takeaways

- Pick $\mathbf{F} = (L, M) \Rightarrow$ 1-form $L\,dx + M\,dy$.
- Circulation (line integral on C) equals total curl over D.
- If $M_x - L_y \equiv 0$ on D, then $\oint_C (L\,dx + M\,dy) = 0$ (path-independence on simply connected D).
- The special 1-form $\frac{1}{2}(x\,dy - y\,dx)$ integrates to the enclosed area.

Problem

$$I = \oint_{\gamma(0,1)} \frac{\Re z}{2z - i}\,dz, \qquad J = \int_0^{2\pi} \frac{\cos^2\theta}{5 - 4\sin\theta}\,d\theta.$$

Evaluate I using only Cauchy's Integral Formula. Then put $z = e^{i\theta}$ on $|z| = 1$ to relate I and J and determine J.

Background idea. $\Re z$ is not holomorphic, but on $|z| = 1$ we may use

$$\Re z = \frac{1}{2}\,(z + \bar{z}) = \frac{1}{2}\,(z + z^{-1}),$$

so

Step 1 — Compute I (Cauchy only)

On $|z| = 1$ write

$$I = \frac{1}{2}\oint_{|z|=1} \frac{z + z^{-1}}{2z - i}\,dz = \frac{1}{2}\left(\underbrace{\oint_{|z|=1} \frac{z}{2z - i}\,dz}_{=:I_1} + \underbrace{\oint_{|z|=1} \frac{1}{z(2z - i)}\,dz}_{=:I_2}\right).$$

First piece.

$$\frac{z}{2z - i} = \frac{1}{2}\left(1 + \frac{i}{2z - i}\right), \qquad \oint_{|z|=1} 1\,dz = 0.$$

Hence

$$I_1 = \frac{i}{2}\oint_{|z|=1} \frac{dz}{2z - i}.$$

There is a simple pole at $z = \dfrac{i}{2}$ inside the unit circle, with residue $\dfrac{1}{2}$. Therefore

$$I_1 = \frac{i}{2} \cdot 2\pi i \cdot \frac{1}{2} = \frac{i \cdot 2\pi i}{4} = -\frac{\pi}{2}.$$

Second piece. Partial fractions:

$$\frac{1}{z(2z-i)} = \frac{A}{z} + \frac{B}{2z-i}, \qquad 1 = A(2z-i) + Bz.$$

Equating coefficients gives $A = i$, $B = -2i$. Thus

$$I_2 = i\oint_{|z|=1} \frac{dz}{z} - 2i\oint_{|z|=1} \frac{dz}{2z-i} = i(2\pi i) - 2i\big(2\pi i \cdot \tfrac{1}{2}\big) = -2\pi + 2\pi = 0.$$

Conclusion for I.

$$I = \frac{1}{2}(I_1 + I_2) = \frac{1}{2}\left(-\frac{\pi}{2} + 0\right) = \boxed{-\frac{\pi}{4}}.$$

Step 2 — Express I as a θ–integral and read off J

On $|z| = 1$ set $z = e^{i\theta}$, $dz = ie^{i\theta}\,d\theta$, and $\Re z = \cos\theta$:

$$I = \int_0^{2\pi} \frac{\cos\theta\; ie^{i\theta}}{2e^{i\theta} - i}\,d\theta.$$

Multiply numerator and denominator by the conjugate along the circle, $2e^{-i\theta} + i$:

$$\frac{ie^{i\theta}}{2e^{i\theta} - i} = \frac{2i - e^{i\theta}}{5 + 4i\sin\theta},$$

so

$$I = \int_0^{2\pi} \frac{\cos\theta\,\big(2i - e^{i\theta}\big)}{5 + 4i\sin\theta}\,d\theta = \int_0^{2\pi} \frac{-\cos^2\theta}{5 - 4\sin\theta}\,d\theta \quad \text{(the imaginary part integrates to 0 over a full period).}$$

Therefore

$$I = -\int_0^{2\pi} \frac{\cos^2\theta}{5 - 4\sin\theta}\,d\theta \;=\; -J.$$

Final values. From Step 1, $I = -\dfrac{\pi}{4}$, hence

$$\boxed{J = \frac{\pi}{4}}.$$

(Optional real-variable check). One may verify directly the identity

$$\frac{\cos^2\theta}{5 - 4\sin\theta} = \frac{5}{16} + \frac{1}{4}\sin\theta - \frac{9}{16}\cdot\frac{1}{5 - 4\sin\theta},$$

and use

$$\int_0^{2\pi} \sin\theta\,d\theta = 0, \qquad \int_0^{2\pi} \frac{d\theta}{a - b\sin\theta} = \frac{2\pi}{\sqrt{a^2 - b^2}} \quad (a > |b|),$$

to obtain $J = \dfrac{5}{16}\cdot 2\pi - \dfrac{9}{16}\cdot\dfrac{2\pi}{\sqrt{25 - 16}} = \dfrac{10\pi}{16} - \dfrac{9\pi}{16} = \dfrac{\pi}{16}\cdot(10 - 9) = \dfrac{\pi}{4}$.

The 1-form and its exterior derivative

Start with the 1-form

$$\alpha \;=\; \frac{1}{2}\,(x\,dy - y\,dx).$$

Rules to use.

$$\begin{aligned}
\text{(Product rule)} \qquad & d(f\,\beta) \;=\; df\wedge\beta \;+\; f\,d\beta,\\
\text{(Nilpotence)} \qquad & d^2 = 0 \;\Rightarrow\; d(dx) = d(dy) = 0,\\
\text{(Antisymmetry)} \qquad & dy\wedge dx \;=\; -\,dx\wedge dy, \quad dx\wedge dx = 0,\; dy\wedge dy = 0.
\end{aligned}$$

Compute $d\alpha$ step by step.

$$\begin{aligned}
d\alpha &= \frac{1}{2}\big(d(x\,dy) - d(y\,dx)\big)\\
&= \frac{1}{2}\Big(dx\wedge dy + x\,d(dy) \;-\; dy\wedge dx - y\,d(dx)\Big) \quad \text{(product rule)}\\
&= \frac{1}{2}\big(dx\wedge dy - dy\wedge dx\big) \qquad\qquad \text{(since } d(dy) = d(dx) = 0\text{)}\\
&= \frac{1}{2}\big(dx\wedge dy - (-dx\wedge dy)\big) \quad \text{(use } dy\wedge dx = -dx\wedge dy\text{)}\\
&= \frac{1}{2}\big(dx\wedge dy + dx\wedge dy\big)\\
&= dx\wedge dy.
\end{aligned}$$

Thus $d\alpha = dx\wedge dy$, the *area 2-form* on the plane.

Stokes/Green gives area from a boundary integral. For a region D with positively oriented boundary ∂D,

$$\oint_{\partial D}\alpha \;=\; \iint_D d\alpha \;=\; \iint_D dx\wedge dy \;=\; \text{Area}(D),$$

i.e.

$$\boxed{\text{Area}(D) \;=\; \frac{1}{2}\oint_{\partial D}\big(x\,dy - y\,dx\big)}\,.$$

Geometric meaning. The 1-form $\alpha = \frac{1}{2}(x\,dy - y\,dx)$ is a symmetric "area-measuring" form on the boundary. The identity $d\alpha = dx\wedge dy$ says its infinitesimal curl equals unit area density. Integrating $d\alpha$ over D gives area; Stokes moves this to the boundary integral of α.

Area via $\frac{1}{2}\oint_{\partial D}(x\,dy - y\,dx)$: Worked Examples

We use

$$\text{Area}(D) = \frac{1}{2}\oint_{\partial D}\big(x\,dy - y\,dx\big).$$

Example 1 — Disk of radius R

Parametrize the boundary circle C:

$$x = R\cos t, \qquad y = R\sin t, \qquad t \in [0, 2\pi],$$

so that

$$dx = -R\sin t\,dt, \qquad dy = R\cos t\,dt.$$

Then

$$\frac{1}{2}\big(x\,dy - y\,dx\big) = \frac{1}{2}\Big(R\cos t \cdot R\cos t - R\sin t \cdot (-R\sin t)\Big)\,dt = \frac{R^2}{2}\,dt.$$

Integrate:

$$\text{Area} = \int_0^{2\pi} \frac{R^2}{2}\,dt = \pi R^2.$$

Example 2 — Triangle with vertices $(0,0) \to (2,1) \to (1,3) \to (0,0)$

Integrate $\frac{1}{2}(x\,dy - y\,dx)$ edge by edge (CCW orientation).

Edge 1: $(0,0) \to (2,1)$**.** Parametrize $x = 2t$, $y = t$, $t \in [0,1]$. Then $dx = 2\,dt$, $dy = dt$, and

$$\frac{1}{2}(x\,dy - y\,dx) = \frac{1}{2}\left(2t \cdot dt - t \cdot 2\,dt\right) = 0 \quad \Rightarrow \quad \text{contribution } = 0.$$

Edge 2: $(2,1) \to (1,3)$**.** Parametrize $x = 2 - s$, $y = 1 + 2s$, $s \in [0,1]$. Then $dx = -ds$, $dy = 2\,ds$, and

$$\frac{1}{2}(x\,dy - y\,dx) = \frac{1}{2}\Big((2-s)\cdot 2 - (1+2s)\cdot(-1)\Big)\,ds = \frac{5}{2}\,ds.$$

Contribution:

$$\int_0^1 \frac{5}{2}\,ds = \frac{5}{2} = 2.5.$$

Edge 3: $(1,3) \to (0,0)$**.** Parametrize $x = 1 - t$, $y = 3 - 3t$, $t \in [0,1]$. Then $dx = -dt$, $dy = -3\,dt$, and

$$\frac{1}{2}(x\,dy - y\,dx) = \frac{1}{2}\Big((1-t)(-3) - (3-3t)(-1)\Big)\,dt = 0 \quad \Rightarrow \quad \text{contribution } = 0.$$

Total area.

$$\text{Area} = 0 + 2.5 + 0 = \boxed{2.5}.$$

(Checks the shoelace formula: $\frac{1}{2}\big|0\cdot 1 + 2\cdot 3 + 1\cdot 0 - (2\cdot 0 + 1\cdot 1 + 0\cdot 3)\big| = \frac{1}{2}|6-1| = 2.5$.)

Problem

Let $\gamma(0,r)$ be the positively oriented circle $|z| = r > 0$. For each integer k, evaluate

$$\oint_{\gamma(0,r)} z^k \, dz,$$

making clear any special cases.

Using $\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$, rewrite

$$\int_0^{2\pi} \sin^{2n}\theta \, d\theta$$

as a contour integral around $\gamma(0,1)$ and deduce

$$\int_0^{2\pi} \sin^{2n}\theta \, d\theta \;=\; \frac{2\pi}{4^n}\binom{2n}{n}.$$

Solution

1) The monomial integral on a circle

Parametrize $z = re^{i\theta}$, $dz = ire^{i\theta}\, d\theta$:

$$\oint_{\gamma(0,r)} z^k \, dz = ir^{k+1}\int_0^{2\pi} e^{i(k+1)\theta}\, d\theta = \begin{cases} 0, & k \neq -1, \\ 2\pi i, & k = -1. \end{cases}$$

Special case: $k = -1$ is the only nonzero value, and it is independent of r.

2) The even power sine integral via a contour integral

On $|z| = 1$ let $z = e^{i\theta}$, so $d\theta = \dfrac{dz}{iz}$ and

$$\sin\theta = \frac{z - 1/z}{2i}.$$

Hence

$$\int_0^{2\pi} \sin^{2n}\theta \, d\theta = \oint_{|z|=1} \frac{(z-1/z)^{2n}}{(2i)^{2n}} \cdot \frac{dz}{iz}.$$

Expand $(z - 1/z)^{2n}$:

$$(z - 1/z)^{2n} = \sum_{m=0}^{2n} \binom{2n}{m} (-1)^m \, z^{2n-2m}.$$

Therefore

$$\frac{(z-1/z)^{2n}}{(2i)^{2n}} \cdot \frac{1}{iz} = \sum_{m=0}^{2n} \binom{2n}{m} \frac{(-1)^m}{(2i)^{2n} i} \, z^{2n-2m-1}.$$

Only the z^{-1} term contributes to the contour integral; this occurs when $2n - 2m - 1 = -1$, i.e. $m = n$. The residue at $z = 0$ is thus

$$\mathrm{Res}_{z=0} = \binom{2n}{n} \frac{(-1)^n}{(2i)^{2n} i}.$$

By the residue theorem,

$$\int_0^{2\pi} \sin^{2n}\theta \, d\theta = 2\pi i \cdot \binom{2n}{n} \frac{(-1)^n}{(2i)^{2n} i} = 2\pi \binom{2n}{n} \frac{(-1)^n}{(2i)^{2n}}.$$

Since $(2i)^{2n} = (4i^2)^n = (-4)^n = (-1)^n 4^n$, the $(-1)^n$ cancels and we obtain

$$\boxed{\int_0^{2\pi} \sin^{2n}\theta \, d\theta = \frac{2\pi}{4^n}\binom{2n}{n}}.$$

Problem

Describe the complex power maps

$$w = z^k, \qquad k = 1, 2, 3, 4,$$

with emphasis on geometry: action on radii/angles, circles/rays, winding, critical points, conformality, and behavior on the unit circle.

Universal facts (all $k \in \mathbb{N}$)

Write $z = re^{i\theta}$ (polar form). Then

$$w = z^k = r^k e^{ik\theta}.$$

Hence, for every k:

- **Radius scaling:** $|w| = |z|^k = r^k$. Circles $|z| = r$ map to circles $|w| = r^k$.
- **Angle multiplication:** $\arg w = k \arg z \pmod{2\pi}$. Rays $\theta = \theta_0$ map to rays $\phi = k\theta_0$.
- **Winding of the unit circle:** $|z| = 1 \mapsto |w| = 1$, wrapped k times (as $\theta : 0 \to 2\pi$ gives $k\theta : 0 \to 2k\pi$).
- **Preimages:** For $w \neq 0$, exactly k distinct preimages (the kth roots of w), spaced by angles $2\pi/k$.
- **Conformality:** Holomorphic and angle-preserving wherever $z \neq 0$.
- **Critical set:** $(z^k)' = kz^{k-1}$ vanishes iff $z = 0$. Thus $z = 0$ is a critical point of multiplicity $k - 1$, locally a $(k : 1)$ folding.
- **Orientation:** Preserved away from 0 (Jacobian $= |kz^{k-1}|^2 > 0$ for $z \neq 0$).

Case-by-case descriptions

$k = 1$ (identity). $w = z$. Radii and angles unchanged. No critical points. Global bijection. One preimage for each w.

$k = 2$ (doubling angle). $w = z^2$. Angles $\phi = 2\theta$; $|w| = r^2$. The unit circle winds twice. Critical behavior: derivative $2z$ vanishes at 0 (local 2:1). Rays map to rays with doubled angle; any line through the origin maps to itself with angle doubled. Real axis ($\theta = 0, \pi$) and imaginary axis ($\theta = \pm\pi/2$) both map to the nonnegative real axis (squaring removes sign).

$k = 3$ (tripling angle). $w = z^3$. Angles $\phi = 3\theta$; $|w| = r^3$. Unit circle wraps three times. Critical: $3z^2$, order-2 at 0 (local 3:1). Preimages are rotated by 120°: if z maps to w, then $ze^{2\pi i/3}$ and $ze^{4\pi i/3}$ do too. Real axis maps to real axis with sign preserved $(\pm x)^3 = \pm x^3$; imaginary axis maps to imaginary axis.

$k = 4$ (quadrupling angle). $w = z^4$. Angles $\phi = 4\theta$; $|w| = r^4$. Unit circle wraps four times. Critical: $4z^3$, order-3 at 0 (local 4:1). Both real and imaginary axes map to the nonnegative real axis (even power discards sign). Preimages exhibit 90° rotational symmetry.

Behavior on the unit circle and line integrals

On $|z| = 1$ one has $z = e^{i\theta} \mapsto w = e^{ik\theta}$: same circle, k-fold wrap. For circle integrals,

$$\oint_{|z|=1} z^k \, dz = \begin{cases} 0, & k \neq -1, \\ 2\pi i, & k = -1, \end{cases}$$

since z^k is holomorphic on and inside the circle for $k \neq -1$, and $\oint dz/z = 2\pi i$.

Visual heuristics

"Multiply the angle, power the radius."

Near 0, z^k collapses many directions into one (derivative 0 at the origin); on the unit circle it is the pure phase map $e^{i\theta} \mapsto e^{ik\theta}$, producing k windings.

Problem

Describe the complex power maps

$$w = z^k, \qquad k = 1, 2, 3, 4,$$

with emphasis on geometry: action on radii/angles, circles/rays, winding, critical points, conformality, and behavior on the unit circle.

Universal facts (all $k \in \mathbb{N}$)

Write $z = re^{i\theta}$ (polar form). Then

$$w = z^k = r^k e^{ik\theta}.$$

Hence, for every k:

- **Radius scaling:** $|w| = |z|^k = r^k$. Circles $|z| = r$ map to circles $|w| = r^k$.
- **Angle multiplication:** $\arg w = k \arg z \pmod{2\pi}$. Rays $\theta = \theta_0$ map to rays $\phi = k\theta_0$.
- **Winding (unit circle):** $|z| = 1 \mapsto |w| = 1$, wrapped k times (as $\theta : 0 \to 2\pi$ gives $k\theta : 0 \to 2k\pi$).
- **Preimages:** For $w \neq 0$, exactly k distinct preimages, spaced by angles $2\pi/k$.
- **Conformality:** Holomorphic and angle-preserving for $z \neq 0$.
- **Critical set:** $(z^k)' = kz^{k-1}$, so $z = 0$ is critical of multiplicity $k - 1$ (local $(k{:}1)$ folding).
- **Orientation:** Preserved away from 0 (Jacobian $= |kz^{k-1}|^2 > 0$ for $z \neq 0$).

Case-by-case (compact, line-by-line)

$\boldsymbol{k = 1}$ **Identity.** $w = z$; radii/angles unchanged; no critical points; bijective; one preimage for each w.

$\boldsymbol{k = 2}$ **Doubling angle.** $w = z^2$; $\phi = 2\theta$, $|w| = r^2$; unit circle winds twice; critical at 0 (order 1; local 2:1); rays $\mapsto$ rays with doubled angle; real/imaginary axes $\mapsto$ nonnegative real axis.

$\boldsymbol{k = 3}$ **Tripling angle.** $w = z^3$; $\phi = 3\theta$, $|w| = r^3$; unit circle winds thrice; critical at 0 (order 2; local 3:1); preimages rotated by 120°; real axis $\mapsto$ real axis, imaginary $\mapsto$ imaginary.

$\boldsymbol{k = 4}$ **Quadrupling angle.** $w = z^4$; $\phi = 4\theta$, $|w| = r^4$; unit circle winds four times; critical at 0 (order 3; local 4:1); real and imaginary axes $\mapsto$ nonnegative real axis; 90° rotational symmetry among preimages.

Behavior on the unit circle and line integrals

On $|z| = 1$ one has $z = e^{i\theta} \mapsto w = e^{ik\theta}$: same circle, k-fold wrap. For circle integrals,

$$\oint_{|z|=1} z^k\,dz = \begin{cases} 0, & k \neq -1, \\ 2\pi i, & k = -1, \end{cases}$$

since z^k is holomorphic on and inside the circle for $k \neq -1$, and $\oint dz/z = 2\pi i$.

Visual heuristic

“Multiply the angle, power the radius.”

Near 0, z^k collapses many directions into one (derivative 0 at the origin); on the unit circle it is the pure phase map $e^{i\theta} \mapsto e^{ik\theta}$, producing k windings.

Case-by-case

$k = 1: \;\; w = z$ (identity)

- **Trivial:** radii and angles unchanged; $|z| = r \mapsto |w| = r, \;\; \theta \mapsto \theta$.
- **No critical points;** bijection; exactly one preimage for each w.

$k = 2: \;\; w = z^2$ (doubling angle)

- **Angles:** $\phi = 2\theta$; the unit circle winds *twice*.
- **Radii:** $|w| = r^2$: the disk $r < 1$ is squeezed ($r \mapsto r^2$), the exterior $r > 1$ expands.
- **Critical behavior:** derivative $2z$ vanishes at 0; locally 2:1.
- **Images of simple sets:**
  - Real axis ($\theta = 0, \pi$) maps to *nonnegative real axis* (both halves merge since $\phi = 2\theta$).
  - Imaginary axis ($\theta = \pm\frac{\pi}{2}$) also maps to *nonnegative real axis.*
  - A line through the origin stays a line through the origin (angle doubled).
- **Fixed points on $|z| = 1$:** $z^2 = z \Rightarrow z \in \{0, 1\}$ (on the circle only $z = 1$).

$k = 3: \;\; w = z^3$ (tripling angle)

- **Angles:** $\phi = 3\theta$; unit circle wraps *three* times.
- **Radii:** $|w| = r^3$.
- **Critical:** derivative $3z^2$, so order-2 critical point at 0; locally 3:1.
- **Symmetry:** $120°$ rotational symmetry in preimages: if z is a preimage, so are $ze^{2\pi i/3}$, $ze^{4\pi i/3}$.
- **Images of axes:**
  - Real axis $\to$ real axis (sign preserved: $(\pm x)^3 = \pm x^3$).
  - Imaginary axis $\to$ imaginary axis: $(iy)^3 = i^3 y^3 = -iy^3$.

$k = 4: \;\; w = z^4$ (quadrupling angle)

- **Angles:** $\phi = 4\theta$; unit circle wraps *four* times.
- **Radii:** $|w| = r^4$.
- **Critical:** derivative $4z^3$, order-3 critical point at 0; locally 4:1.
- **Axes mapping:** both real and imaginary axes $\mapsto$ *nonnegative real axis* (even power removes sign).
- **Symmetry:** $90°$ rotational symmetry among preimages.

Problem

Let $a \in \mathbb{R}$ with $a > 1$. Using Cauchy's Integral Formula (residue theorem) and the holomorphic function

$$f(z) = \frac{z^n}{z-a},$$

compute

$$I_n(a) := \int_0^{2\pi} \frac{\cos(n\theta)}{1 - 2a\cos\theta + a^2}\, d\theta.$$

Setup: unit circle substitution

On $|z| = 1$ let $z = e^{i\theta}$, so $d\theta = \dfrac{dz}{iz}$ and

$$\cos(n\theta) = \tfrac{1}{2}\left(z^n + z^{-n}\right), \qquad 1 - 2a\cos\theta + a^2 = \frac{(z-a)(1-az)}{z}.$$

Hence

$$I_n(a) = \frac{1}{2i} \oint_{|z|=1} \frac{z^n + z^{-n}}{(z-a)(1-az)}\, dz.$$

Write

$$g(z) := \frac{z^n + z^{-n}}{(z-a)(1-az)}.$$

For $a > 1$, the poles of g *inside* $|z| = 1$ are:

$$z = \frac{1}{a} \quad \text{(simple)}, \qquad z = 0 \text{ (present only when } n \geq 1).$$

(The pole at $z = a$ lies outside.)

By the residue theorem,

$$I_n(a) = \frac{1}{2i} \cdot 2\pi i \sum_{\text{inside}} \operatorname{Res}(g) = \pi \sum_{\text{inside}} \operatorname{Res}(g).$$

Residues

At $z = \frac{1}{a}$. Since $1 - az$ vanishes simply,

$$\operatorname{Res}_{z=1/a} g = \lim_{z \to 1/a} \frac{z^n + z^{-n}}{(z-a)} \cdot \frac{1}{-a} = \frac{a^{-n} + a^n}{a^2 - 1}.$$

At $z = 0$ (only if $n \geq 1$). Use the double geometric series

$$\frac{1}{(z-a)(1-az)} = -\frac{1}{a}\,\frac{1}{1-z/a}\,\frac{1}{1-az} = -\frac{1}{a} \sum_{m \geq 0} \left(\frac{z}{a}\right)^m \sum_{\ell \geq 0} (az)^\ell.$$

Multiplying by z^{-n}, the coefficient of z^{-1} (the residue) arises from $m + \ell = n - 1$:

$$\operatorname{Res}_{z=0} g = -\frac{1}{a} \sum_{m=0}^{n-1} a^{\ell - m} = -\frac{1}{a} \sum_{m=0}^{n-1} a^{n-1-2m} = -\frac{a^n - a^{-n}}{a^2 - 1}.$$

Sum and value of the integral

For $n \geq 1$,

$$\sum \operatorname{Res}(g) = \frac{a^{-n} + a^{n}}{a^2 - 1} - \frac{a^{n} - a^{-n}}{a^2 - 1} = \frac{2a^{-n}}{a^2 - 1}.$$

Therefore

$$\boxed{I_n(a) = \pi \cdot \frac{2a^{-n}}{a^2 - 1} = \frac{2\pi}{a^2 - 1}\, a^{-n}, \qquad a > 1,\ n \geq 1.}$$

For $n = 0$ there is no pole at 0, so

$$\boxed{I_0(a) = \pi \cdot \frac{2}{a^2 - 1} = \frac{2\pi}{a^2 - 1}.}$$

Checks

$$I_0(a) = \int_0^{2\pi} \frac{d\theta}{1 - 2a\cos\theta + a^2} = \frac{2\pi}{a^2 - 1}, \qquad I_1(a) = \frac{2\pi}{a^2 - 1} \cdot \frac{1}{a}, \quad I_2(a) = \frac{2\pi}{a^2 - 1} \cdot \frac{1}{a^2}, \ \ldots$$

These match the Poisson–kernel/Fourier identity $\dfrac{1}{1 - 2a\cos\theta + a^2} = \dfrac{1}{a^2 - 1}\Big(1+2\displaystyle\sum_{m\geq 1} a^{-m}\cos m\theta\Big)$ for $a > 1$.

Problem

For $a > 1$ and an integer $k \geq 0$, evaluate

$$I_k(a) := \int_0^{2\pi} \frac{\cos(k\theta)}{1 - 2a\cos\theta + a^2}\, d\theta,$$

using Cauchy's Integral Formula with the holomorphic function

$$f(z) = \frac{z^{k}}{z - a}.$$

Then give the explicit values for $k = 1, 2, 3, 4$ and interpret the pattern.

Master formula (from the contour proof)

Let $z = e^{i\theta}$ on $|z| = 1$, so $d\theta = \dfrac{dz}{iz}$, $\cos(k\theta) = \frac{1}{2}(z^k + z^{-k})$, and

$$1 - 2a\cos\theta + a^2 = \frac{(z - a)(1 - az)}{z}.$$

Hence

$$I_k(a) = \frac{1}{2i} \oint_{|z|=1} \frac{z^{k} + z^{-k}}{(z - a)(1 - az)}\, dz.$$

For $a > 1$, the poles inside $|z| = 1$ are $z = \frac{1}{a}$ (simple) and, when $k \geq 1$, also $z = 0$ from z^{-k}. A short residue computation gives

$$\sum_{\text{inside}} \operatorname{Res}\left(\frac{z^k + z^{-k}}{(z-a)(1-az)}\right) = \frac{2a^{-k}}{a^2-1},$$

so

$$\boxed{I_k(a) = \frac{2\pi}{a^2-1}\, a^{-k}, \qquad a > 1,\ k \in \mathbb{N}.}$$

For $k = 0$ there is no pole at 0, and we get

$$\boxed{I_0(a) = \frac{2\pi}{a^2-1}.}$$

Explicit values for $k = 1, 2, 3, 4$

$$\begin{aligned} I_1(a) &= \frac{2\pi}{a^2-1} \cdot \frac{1}{a}, \\ I_2(a) &= \frac{2\pi}{a^2-1} \cdot \frac{1}{a^2}, \\ I_3(a) &= \frac{2\pi}{a^2-1} \cdot \frac{1}{a^3}, \\ I_4(a) &= \frac{2\pi}{a^2-1} \cdot \frac{1}{a^4}. \end{aligned}$$

Why the pattern is so clean

(1) Poisson kernel / Fourier viewpoint

Set $r = \frac{1}{a} \in (0, 1)$. Then

$$\frac{1}{1 - 2a\cos\theta + a^2} = \frac{1}{a^2} \cdot \frac{1}{1 - 2r\cos\theta + r^2} = \frac{1}{a^2-1}\Big(1 + 2\sum_{m\geq 1} r^m \cos(m\theta)\Big).$$

Multiply by $\cos(k\theta)$ and integrate over $[0, 2\pi]$; orthogonality leaves only the $m = k$ term:

$$I_k(a) = \frac{1}{a^2-1} \cdot 2r^k \int_0^{2\pi} \cos^2(k\theta)\, d\theta = \frac{2\pi}{a^2-1}\, r^k = \frac{2\pi}{a^2-1}\, a^{-k}.$$

For $k = 0$ only the constant term contributes, giving $I_0(a) = \frac{2\pi}{a^2-1}$.

(2) Contour / residue viewpoint

With $z = e^{i\theta}$,

$$I_k(a) = \frac{1}{2i} \oint_{|z|=1} \frac{z^k + z^{-k}}{(z-a)(1-az)}\, dz,$$

and the only interior poles are at $z = 1/a$ and (if $k \geq 1$) $z = 0$. Their residues sum to $\dfrac{2a^{-k}}{a^2-1}$, so $I_k(a) = \dfrac{2\pi}{a^2-1}\, a^{-k}$ again.

Interpretation as k varies

- **Exponential decay in k:** $I_k(a) \sim C\, a^{-k}$—higher harmonics are damped as the parameter $a > 1$ pushes the pole farther from the unit circle.
- **Monotone in a:** For fixed $k \geq 1$, $I_k(a)$ decreases with a.
- **Base case $k = 0$:** Recovers the classical identity $\displaystyle\int_0^{2\pi} \frac{d\theta}{a^2 - 2a\cos\theta + 1} = \frac{2\pi}{a^2 - 1}$.

Setup

Fix $a \in \mathbb{R}$ with $a > 1$. For $n \in \mathbb{N}$, consider

$$F_n(z) = \frac{z^n}{z - a}.$$

We record zeros/poles, partial fraction form, residue at a, behavior at ∞, critical points, and a brief geometric note.

Shared facts (all $n \geq 1$)

- **Meromorphic on $\widehat{\mathbb{C}}$:** a simple pole at $z = a$; a zero of order n at $z = 0$.
- **Residue at a:**
$$\mathrm{Res}_{z=a}\, F_n = \lim_{z \to a} (z - a) \frac{z^n}{z - a} = a^n.$$
- **Polynomial part (long division):**
$$\frac{z^n}{z - a} = z^{n-1} + az^{n-2} + \cdots + a^{n-2}z + a^{n-1} \; + \; \frac{a^n}{z - a}.$$
- **Growth at ∞:** $F_n(z) \sim z^{n-1}$ as $|z| \to \infty$. Thus ∞ is a critical point of multiplicity $n - 2$ when $n \geq 2$ (removable for $n = 1$).
- **Derivative / finite critical points:**
$$F_n'(z) = \frac{z^{n-1}\big((n-1)z - na\big)}{(z - a)^2}.$$

  Hence, aside from the pole, the finite critical points are

$$z = 0 \text{ (multiplicity } n - 1 \text{ if } n \geq 2), \qquad z = \frac{n}{n-1} a \text{ (simple, for } n \geq 2).$$

  (This matches Riemann–Hurwitz: total branching $= 2n - 2$.)

Case-by-case

$n = 1: \ F_1(z) = \dfrac{z}{z-a}$ **(Möbius).**

- Zeros/poles: simple zero at 0; simple pole at a.
- Residue: $\operatorname{Res}_a F_1 = a$.
- Partial fraction: $F_1(z) = 1 + \dfrac{a}{z-a}$.
- Growth: $F_1(\infty) = 1$ (removable at ∞).
- Critical points: none in $\mathbb{C}$ ($F_1'(z) = -a/(z-a)^2 \neq 0$).
- Geometry: Möbius map; sends lines/circles to lines/circles; conformal except at $z = a$.

$n = 2: \ F_2(z) = \dfrac{z^2}{z-a}$.

- Zeros/poles: double zero at 0; simple pole at a.
- Residue: $\operatorname{Res}_a F_2 = a^2$.
- Partial fraction: $F_2(z) = z + a + \dfrac{a^2}{z-a}$.
- Growth: $F_2(z) \sim z$ as $|z| \to \infty$ (simple pole at ∞).
- Critical points from $F_2'(z) = \dfrac{z(2z-2a)}{(z-a)^2}$: $z = 0$ (order 1), $z = 2a$ (simple).
- Geometry: degree-2 rational map (generically 2:1); near a, behaves like $a^2/(z-a)$.

$n = 3: \ F_3(z) = \dfrac{z^3}{z-a}$.

- Zeros/poles: triple zero at 0; simple pole at a.
- Residue: $\operatorname{Res}_a F_3 = a^3$.
- Partial fraction: $F_3(z) = z^2 + az + a^2 + \dfrac{a^3}{z-a}$.
- Growth: $F_3(z) \sim z^2$ (double pole at ∞).
- Critical points from $F_3'(z) = \dfrac{z^2(2z-3a)}{(z-a)^2}$: $z = 0$ (order 2), $z = \frac{3}{2}a$ (simple); plus one at ∞ (order 1), totaling $4 = 2n - 2$.
- Geometry: degree-3 branched covering; a generic value has three preimages (with multiplicity).

$n = 4: \; F_4(z) = \dfrac{z^4}{z-a}$.

- Zeros/poles: quadruple zero at 0; simple pole at a.
- Residue: $\operatorname{Res}_a F_4 = a^4$.
- Partial fraction: $F_4(z) = z^3 + az^2 + a^2 z + a^3 + \dfrac{a^4}{z-a}$.
- Growth: $F_4(z) \sim z^3$ (triple pole at ∞).
- Critical points from $F_4'(z) = \dfrac{z^3(3z-4a)}{(z-a)^2}$: $z = 0$ (order 3), $z = \frac{4}{3}a$ (simple); plus a critical point at ∞ (order 2); total branching $6 = 2n - 2$.
- Geometry: degree-4 map; generic value has four preimages; stronger "folding" near 0.

Unit-circle snapshot $(|z| = 1,\ a > 1)$

$$F_n(e^{it}) = \frac{e^{int}}{e^{it} - a}.$$

The denominator never vanishes on $|z| = 1$; the magnitude $|F_n(e^{it})| = 1/|e^{it} - a|$ is independent of n, while the phase winds n times as $t : 0 \to 2\pi$. This is exactly why, in integrals against $\cos(n\theta)$, a clean factor a^{-n} appears.

Problem

Consider the oriented closed curve γ shown. Compute the winding number

$$\operatorname{Ind}(\gamma; p) = \frac{1}{2\pi i} \oint_\gamma \frac{dz}{z - p}$$

around each marked point $p \in \{a, b, c, d, e\}$.

Background: what the winding number is and how to read it

- **Meaning.** $\operatorname{Ind}(\gamma; p)$ counts, with sign, how many times γ winds around p: $+1$ for a counterclockwise (CCW) loop, -1 for a clockwise (CW) loop, and sums over multiple loops.
- **Ray test (practical rule).** Draw a ray from p to infinity (e.g. horizontally to the right). Count intersections of γ with the ray: each upward crossing contributes $+1$, each downward crossing contributes -1; the sum equals $\operatorname{Ind}(\gamma; p)$. Any generic ray avoiding vertices gives the same integer.
- **Reading orientation on a small lobe.**
  - On a *CCW* lobe: right edge arrows go up; along the top they go left; left edge down; bottom right.
  - On a *CW* lobe: right edge arrows go down; along the top they go right; left edge up; bottom left.

Reading the figure (qualitative description)

- Point a lies inside a right-hand lobe with arrows up on the right edge and left along the top: a CCW loop. Hence $\operatorname{Ind}(\gamma; a) = +1$.
- Point b lies in a smaller right–middle lobe that is CCW. Hence $\operatorname{Ind}(\gamma; b) = +1$.
- Point c lies in a left interior oval whose overall orientation is clockwise. Hence $\operatorname{Ind}(\gamma; c) = -1$.
- Point d is not encircled by any lobe (strands pass on both sides with no net wrap). Hence $\operatorname{Ind}(\gamma; d) = 0$.
- Point e is inside a bottom oval oriented clockwise (top edge arrows run right). Hence $\operatorname{Ind}(\gamma; e) = -1$.

Answer (collected)

$$\boxed{\operatorname{Ind}(\gamma; a) = +1, \qquad \operatorname{Ind}(\gamma; b) = +1, \qquad \operatorname{Ind}(\gamma; c) = -1, \qquad \operatorname{Ind}(\gamma; d) = 0, \qquad \operatorname{Ind}(\gamma; e) = -1.}$$

Problem

Give a compact profile of the meromorphic function

$$F_p(z) = \frac{1}{z-p} \qquad (p \in \mathbb{C}),$$

and explain why the winding number

$$\operatorname{Ind}(\gamma; p) = \frac{1}{2\pi i} \oint_\gamma \frac{dz}{z-p}$$

is constant on connected components of $\mathbb{C} \setminus \gamma$.

The function $F_p(z) = 1/(z-p)$

Algebraic/analytic data

- **Domain.** Holomorphic on $\mathbb{C} \setminus \{p\}$.
- **Zeros.** None in $\mathbb{C}$; on $\widehat{\mathbb{C}}$ there is a simple zero at ∞.
- **Pole.** Simple pole at $z = p$ with principal part $\dfrac{1}{z-p}$.
- **Residue.** $\operatorname{Res}_{z=p} F_p = 1$.
- **Laurent at p.** $\dfrac{1}{z-p}$ (pure principal part).

- **Power series at $z_0 \neq p$.** If $|z - z_0| < |z_0 - p|$, then

$$\frac{1}{z-p} = \frac{1}{z_0 - p} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{z_0 - p} \right)^n .$$

- **Derivatives.** $F_p'(z) = -\dfrac{1}{(z-p)^2}$, $\quad F_p''(z) = \dfrac{2}{(z-p)^3}$, ... Hence no finite critical points (derivative never 0); branching only at ∞.

Geometry / mapping properties (Möbius transform)

- **Decomposition.** $F_p = \big(z \mapsto 1/z\big) \circ \big(z \mapsto z - p\big)$.
- **Conformality.** Holomorphic and locally orientation-preserving on $\mathbb{C} \setminus \{p\}$.
- **Circles/lines.** Möbius maps send generalized circles to generalized circles. Any circle/line passing through p maps to a line; otherwise to a circle not through 0.
- **Polar description.** If $z - p = \rho e^{i\theta}$, then

$$w = F_p(z) = \frac{1}{\rho} e^{-i\theta}.$$

  Radii invert ($\rho \mapsto 1/\rho$) and angles negate ($\theta \mapsto -\theta$). The circle $|z - p| = \rho$ maps to $|w| = 1/\rho$.

Behavior at ∞

As $|z| \to \infty$, $F_p(z) \to 0$. On $\widehat{\mathbb{C}}$, $w = 0$ is the value at ∞ and F_p has a simple zero at ∞. Being a degree-1 rational map, F_p is a biholomorphism of the sphere.

Why $\mathrm{Ind}(\gamma; p)$ is constant on components of $\mathbb{C} \setminus \gamma$

(A) Homotopy invariance (Cauchy)

If γ_0, γ_1 are closed curves homotopic in $\mathbb{C} \setminus \{p\}$, then by Cauchy's theorem

$$\oint_{\gamma_0} \frac{dz}{z-p} = \oint_{\gamma_1} \frac{dz}{z-p},$$

so $\mathrm{Ind}(\gamma; p)$ depends only on the homotopy class of γ in $\mathbb{C} \setminus \{p\}$. Hence $p \mapsto \mathrm{Ind}(\gamma; p)$ is locally constant on each connected component of $\mathbb{C} \setminus \gamma$.

(B) Argument–change formula

Choose a branch of $\mathrm{Log}(z - p)$ along γ. Since $\dfrac{d}{dz} \mathrm{Log}(z-p) = \dfrac{1}{z-p}$,

$$\oint_{\gamma} \frac{dz}{z-p} = \oint_{\gamma} d\, \mathrm{Log}(z-p) = \Delta_\gamma \mathrm{Log}(z-p) = i\, \Delta_\gamma \mathrm{Arg}(z-p),$$

and therefore

$$\boxed{\operatorname{Ind}(\gamma;p)=\frac{1}{2\pi}\,\Delta_\gamma \operatorname{Arg}\big(\gamma(t)-p\big)}\,.$$

This is the rigorous form of the "ray test" for counting signed crossings.

(C) Via the map $w = 1/(z-p)$

Under $w = F_p(z)$ we have Arg $w = -\operatorname{Arg}(z-p)$, hence

$$\Delta_\gamma \operatorname{Arg}(z-p) = -\,\Delta_{w(\gamma)} \operatorname{Arg}(w) \quad\Rightarrow\quad \oint_\gamma \frac{dz}{z-p} = -\oint_{w(\gamma)} \frac{dw}{w} = -\,2\pi i \operatorname{Ind}\big(w(\gamma);0\big) = 2\pi i \operatorname{Ind}(\gamma;p).$$

Thus the index is the signed number of turns of $w(\gamma)$ about 0.

TL;DR

$$F_p(z) = \frac{1}{z-p} : \text{ meromorphic, simple pole at } p \text{ (Res} = 1), \text{ simple zero at } \infty,$$
$$F_p'(z) = -\frac{1}{(z-p)^2}, \text{ Möbius (translation+inversion), circles/lines } \to \text{ circles/lines.}$$

Along a loop around p: radius inverts, angle negates; the total change in $\operatorname{Arg}(z-p)$ equals $2\pi \cdot \operatorname{Ind}$. Hence $\operatorname{Ind}(\gamma;p)$ is constant on components of $\mathbb{C}\setminus\gamma$.

Problem 1 *Let $f(z) = e^{-z^2}$. Using an appropriate contour and the Cauchy–Goursat theorem, show that for $b > 0$,*

$$\int_0^\infty e^{-x^2}\cos(2bx)\,dx = \frac{\sqrt{\pi}}{2}\,e^{-b^2}.$$

You may use $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$.

Solution 1 ***Step 1: The rectangle.*** *Consider the rectangle with vertices $-R$, R, $R+ib$, $-R+ib$ oriented counterclockwise. Since e^{-z^2} is entire, Cauchy–Goursat gives*

$$\int_{-R}^{R} e^{-x^2}\,dx + \int_{right} e^{-z^2}\,dz - \int_{-R}^{R} e^{-(x+ib)^2}\,dx + \int_{left} e^{-z^2}\,dz = 0.$$

Step 2: Vertical sides vanish. *On $z = \pm R + iy$ with $0 \le y \le b$,*

$$|e^{-z^2}| = e^{-(R^2-y^2)} \le e^{-R^2}e^{b^2},$$

so both vertical integrals tend to 0 as $R \to \infty$.

Step 3: Shifted top edge. *Letting $R \to \infty$ yields*

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \int_{-\infty}^{\infty} e^{-(x+ib)^2}\,dx.$$

Since $e^{-(x+ib)^2} = e^{-x^2+b^2-i\,2bx} = e^{b^2}\,e^{-x^2}\,e^{-i2bx}$*, we get*

$$\int_{-\infty}^{\infty} e^{-x^2} e^{\,i2bx}\,dx = e^{-b^2}\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}\,e^{-b^2}.$$

Step 4: Take real parts and halve. *Because the integrand is even,*

$$\begin{aligned}\int_{0}^{\infty} e^{-x^2}\cos(2bx)\,dx &= \frac{1}{2}\int_{-\infty}^{\infty} e^{-x^2}\cos(2bx)\,dx\\ &= \frac{1}{2}\,\Re\Big(\int_{-\infty}^{\infty} e^{-x^2} e^{\,i2bx}\,dx\Big)\\ &= \frac{1}{2}\,\Re\big(\sqrt{\pi}\,e^{-b^2}\big) = \frac{\sqrt{\pi}}{2}\,e^{-b^2}.\end{aligned}$$

□

Background on the function $f(z) = e^{-z^2}$

Domain and analyticity. The function $f(z) = e^{-z^2}$ is entire (holomorphic on all of $\mathbb{C}$) with global power series

$$e^{-z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!}\,z^{2n}.$$

Derivative, critical set, and ODE. We have

$$f'(z) = -2z\,e^{-z^2} = -2z\,f(z),$$

so the unique critical point is $z = 0$ (and the corresponding critical value is $f(0) = 1$). The function solves the first-order linear ODE $f'(z) + 2z\,f(z) = 0$.

Zeros and singularities. The exponential never vanishes, so f has no zeros in $\mathbb{C}$. Being entire, it has no poles; its only singularity is the essential singularity at infinity. Its growth is anisotropic: for $z = re^{i\theta}$,

$$|f(z)| = e^{-\operatorname{Re}(z^2)} = e^{-r^2\cos(2\theta)},$$

which decays when $|\theta| < \pi/4$ and explodes when $\pi/4 < |\theta| < 3\pi/4$.

Symmetries. Since the Taylor coefficients are real, $f(\overline{z}) = \overline{f(z)}$. Moreover f is even: $f(-z) = f(z)$.

Modulus and phase; characteristic curves. Writing $z = x + iy$,

$$e^{-z^2} = e^{-(x^2-y^2)}\,e^{-i\,2xy}.$$

Hence

$$|f(z)| = e^{-(x^2-y^2)}, \qquad \arg f(z) = -2xy.$$

Therefore:

- Level sets of constant modulus $|f| = \rho$ are the hyperbolas $x^2 - y^2 = -\log \rho$ (rectangular hyperbolas with asymptotes $y = \pm x$).
- Level sets of constant argument $\arg f = \theta$ are the hyperbolas $xy = -\frac{\theta}{2}$, orthogonal to the modulus curves.

On the diagonals $y = \pm x$ we have $|f| = 1$ and $\arg f = \mp 2x^2$. On the real axis $f(x) = e^{-x^2} \in (0, 1]$; on the imaginary axis $f(iy) = e^{y^2} > 1$.

Conformality. The map is conformal everywhere except at $z = 0$, where $f'(0) = 0$.

Order and Fourier connection. The function is an entire function of order 2. The Gaussian is (up to normalization) an eigenfunction of the Fourier transform:

$$\int_{-\infty}^{\infty} e^{-x^2} e^{i2bx}\, dx = \sqrt{\pi}\, e^{-b^2},$$

and the standard rectangle-shift contour argument provides a classical complex-analytic proof.

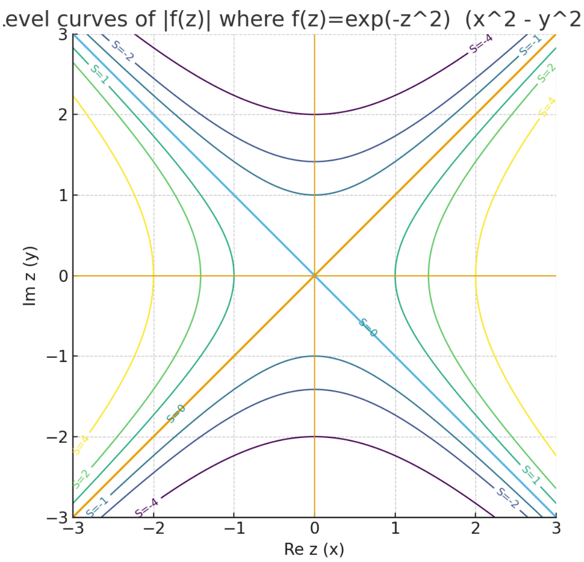


Figure 1: Level curves of $|f(z)|$ for $f(z) = e^{-z^2}$. The curves are $x^2 - y^2 = S$ (rectangular hyperbolas; asymptotes $y = \pm x$). Along $|\arg z| < \pi/4$ the modulus decays; near the imaginary axis it grows.

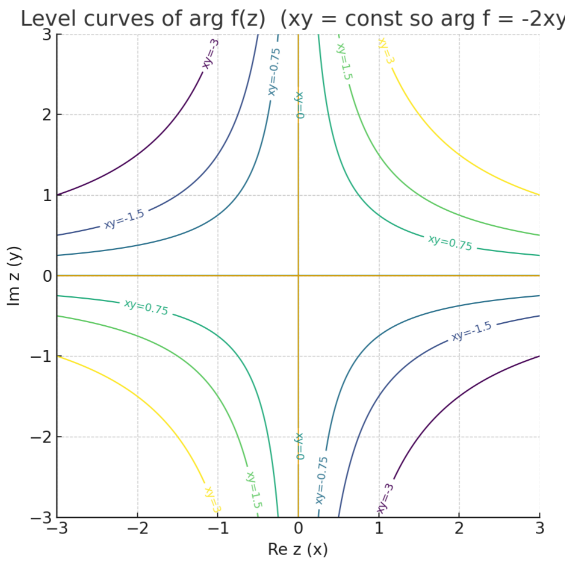
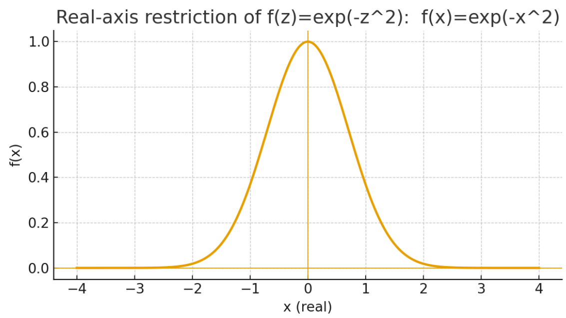


Figure 2: Level curves of $\arg f(z)$. They are $xy = \text{const}$, orthogonal to the modulus curves. On the diagonals $y = \pm x$ one has $|f| = 1$ and $\arg f = \mp 2x^2$.

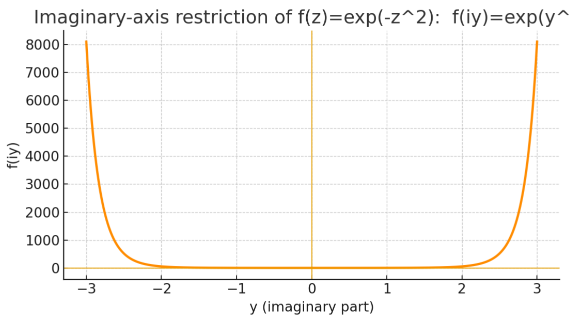
Background on $g_b(x) = e^{-x^2} \cos(2bx)$

Domain and smoothness. g_b is defined and C^∞ on all of $\mathbb{R}$ (and entire as a function of $x \in \mathbb{C}$).

Symmetry. g_b is even: $g_b(-x) = g_b(x)$.



(a) Real-axis restriction $f(x) = e^{-x^2}$.



(b) Imaginary-axis restriction $f(iy) = e^{y^2}$.

Figure 3: Restrictions of $f(z) = e^{-z^2}$ on the real and imaginary axes.

Envelope and oscillation. $|g_b(x)| \le e^{-x^2}$, so the Gaussian envelope controls the size. Oscillation frequency scales linearly with b; as $b \to 0$, $g_b \to e^{-x^2}$.

Zeros and critical points. Zeros occur when $\cos(2bx) = 0$, i.e.

$$x = \frac{(k + \frac{1}{2})\pi}{2b}, \qquad k \in \mathbb{Z},\ b > 0.$$

Differentiating,

$$g_b'(x) = e^{-x^2}\big(- 2x\cos(2bx) - 2b\sin(2bx)\big) = 0 \quad \iff \quad \tan(2bx) = -\frac{x}{b},$$

so local extrema lie near $x \approx \frac{k\pi}{2b}$, shifted slightly toward the origin.

Fourier/Gaussian connection. With $h_b(x) = e^{-x^2} e^{i2bx}$, one has $g_b(x) = \Re h_b(x)$ and

$$\int_{-\infty}^{\infty} h_b(x)\, dx = \sqrt{\pi}\, e^{-b^2},$$

whence

$$\int_0^{\infty} e^{-x^2} \cos(2bx)\, dx = \frac{\sqrt{\pi}}{2}\, e^{-b^2} \quad \text{(even integrand)}.$$

Integrability and L^2 size. $g_b \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, and

$$\int_{-\infty}^{\infty} g_b(x)\, dx = \sqrt{\pi}\, e^{-b^2}, \qquad \int_{-\infty}^{\infty} g_b(x)^2\, dx = \frac{\sqrt{\pi/2}}{2}\,\big(1 + e^{-2b^2}\big).$$

Differential relation (via complex parent). For $h_b(x) = e^{-x^2} e^{i2bx}$ one has $h_b'(x) = (-2x + i2b)h_b(x)$ and $g_b = \Re h_b$.

Theory notes: $\sinh x$ and $\cos x$ (with plots)

Definitions and core identities

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cos x = \frac{e^{ix} + e^{-ix}}{2} = \Re\big(e^{ix}\big)\,.$$

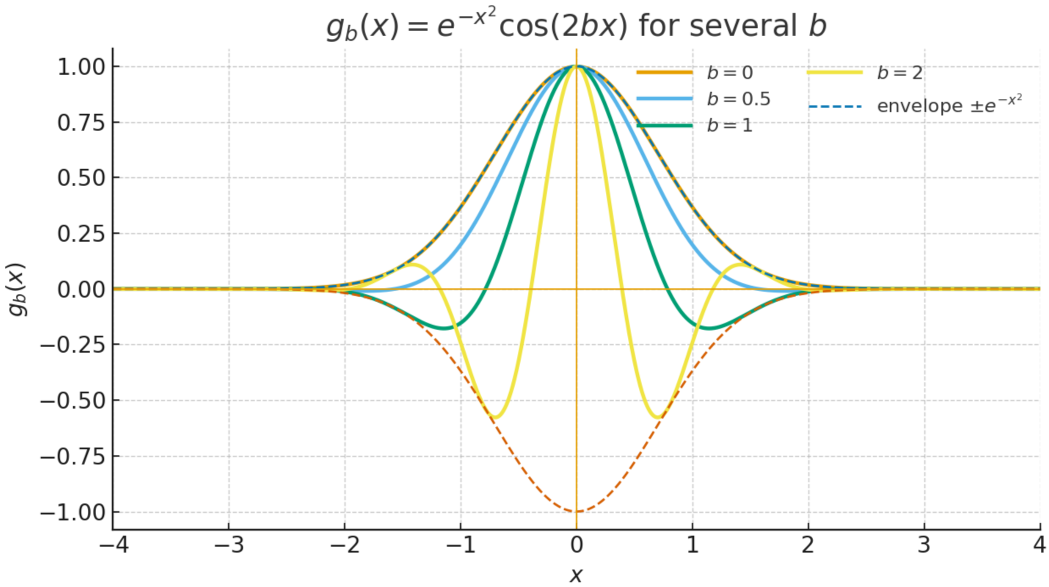
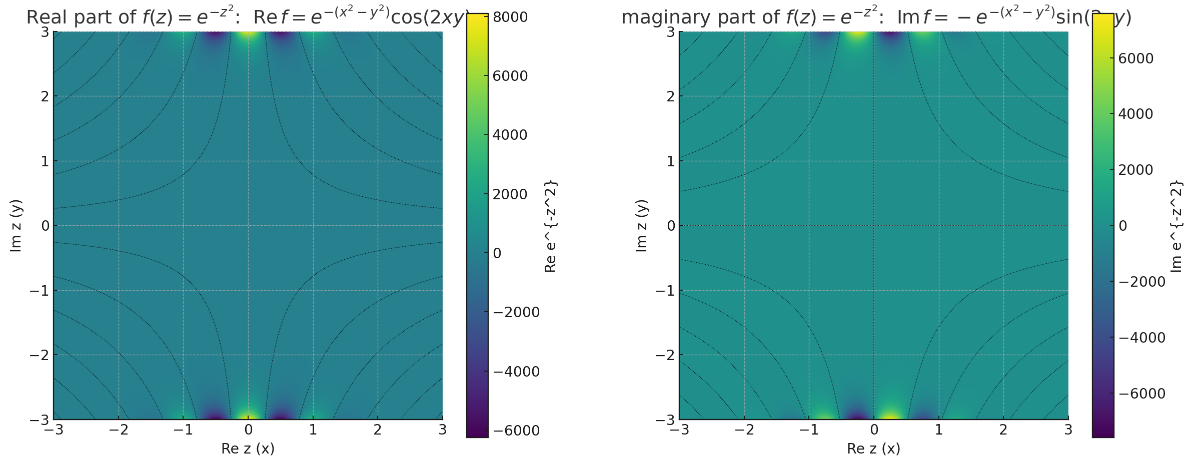


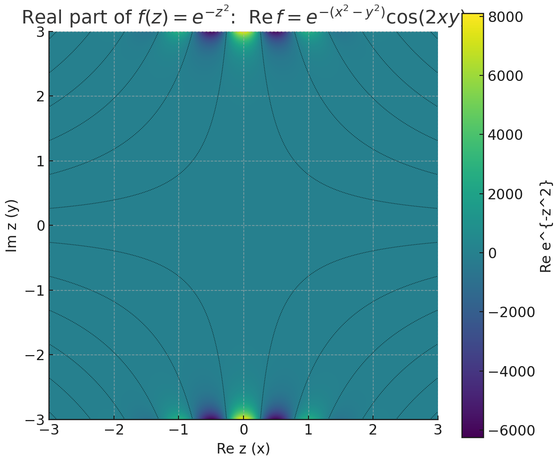
Figure 4: The functions $g_b(x) = e^{-x^2}\cos(2bx)$ for $b \in \{0,\ 0.5,\ 1,\ 2\}$. The dashed curves show the envelope $\pm e^{-x^2}$. As b increases, the oscillation frequency grows while the Gaussian envelope continues to control the magnitude.



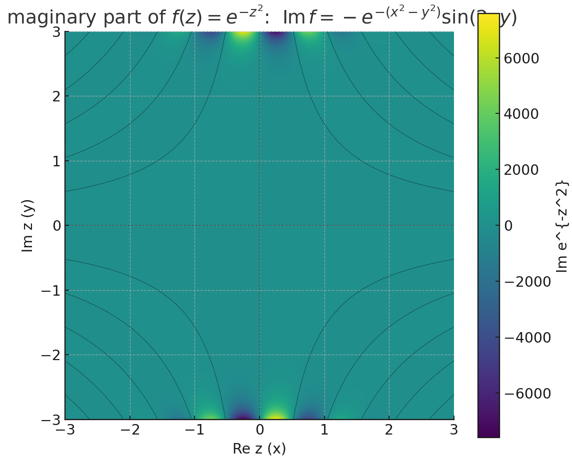
(a) Real part $\mathrm{Re}\, f = e^{-(x^2-y^2)}\cos(2xy)$ with nodal set $xy = \frac{\pi}{4}(2k+1)$.

(b) Imaginary part $\mathrm{Im}\, f = -e^{-(x^2-y^2)}\sin(2xy)$ with nodal set $xy = \frac{k\pi}{2}$.

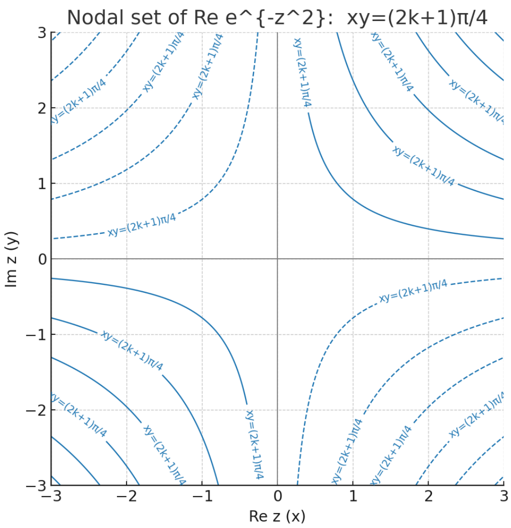
Figure 5: Components of $f(z) = e^{-z^2}$ over the $z = x + iy$ plane. Hyperbola families for modulus/phase appear as nodal sets of $\Re f$ and $\Im f$.



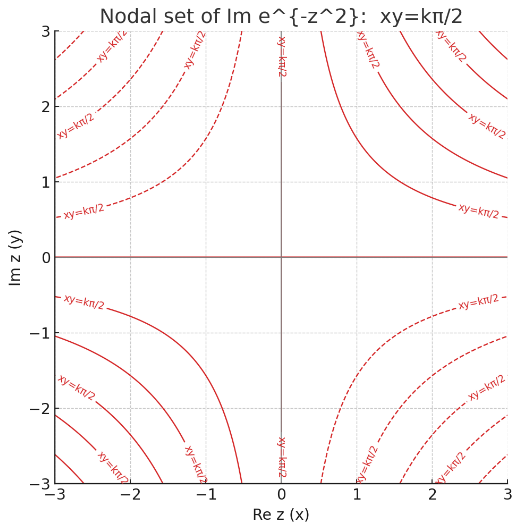
(a) Real part $\mathrm{Re}\, f = e^{-(x^2-y^2)} \cos(2xy)$.
Hyperbola nodal set at $xy = \frac{\pi}{4}(2k+1)$.



(b) Imaginary part $\mathrm{Im}\, f = -e^{-(x^2-y^2)} \sin(2xy)$.
Hyperbola nodal set at $xy = \frac{k\pi}{2}$.



(c) Nodal hyperbolas of $\mathrm{Re}\, f$: $xy = \frac{\pi}{4}(2k+1)$.



(d) Nodal hyperbolas of $\mathrm{Im}\, f$: $xy = \frac{k\pi}{2}$.

Figure 6: Components of $f(z) = e^{-z^2}$ over $z = x + iy$. Top row: heatmaps of $\mathrm{Re}\, f$ and $\mathrm{Im}\, f$. Bottom row: their interleaved nodal sets (same orientation, offset by a $\pi/4$ shift).

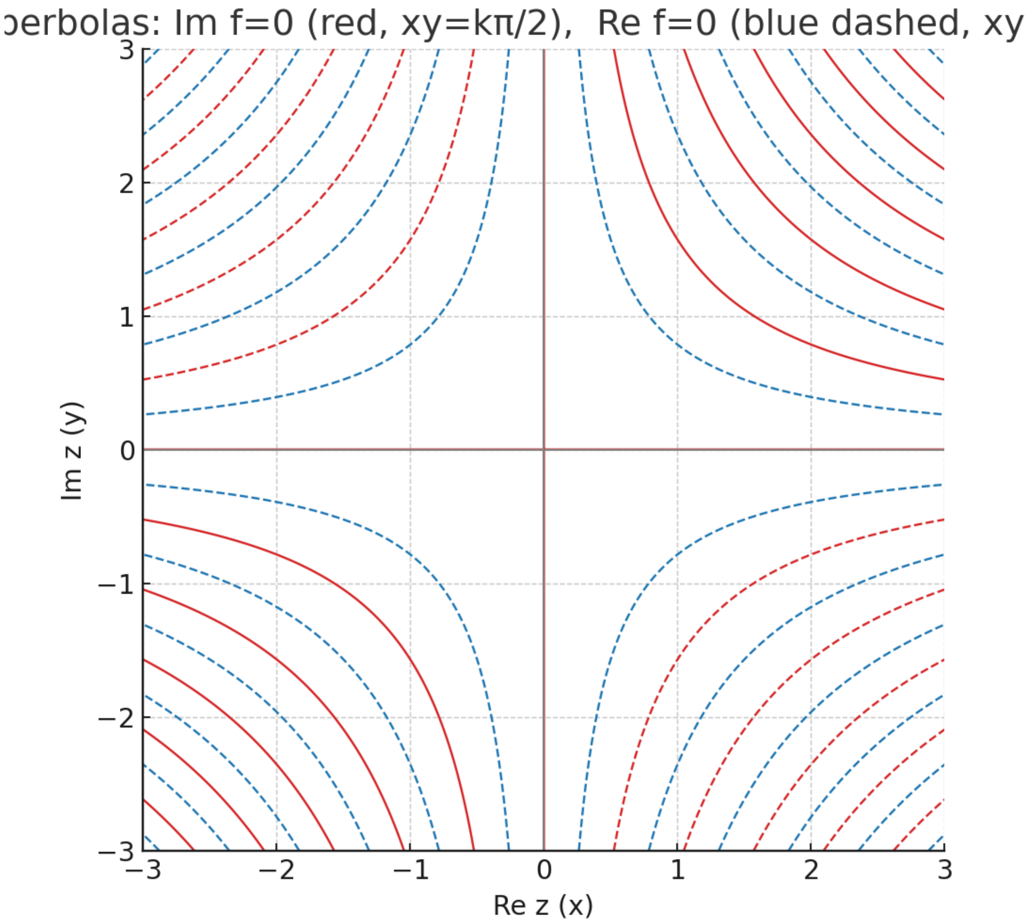


Figure 7: Nodal hyperbolas overlay: $\operatorname{Im} f = 0$ (red, $xy = k\pi/2$) and $\operatorname{Re} f = 0$ (blue dashed, $xy = (2k+1)\pi/4$).

Useful complex links:

$$\sinh(ix) = i\sin x, \qquad \cosh(ix) = \cos x.$$

Pythagorean-type identity (hyperbolic):

$$\cosh^2 x - \sinh^2 x = 1.$$

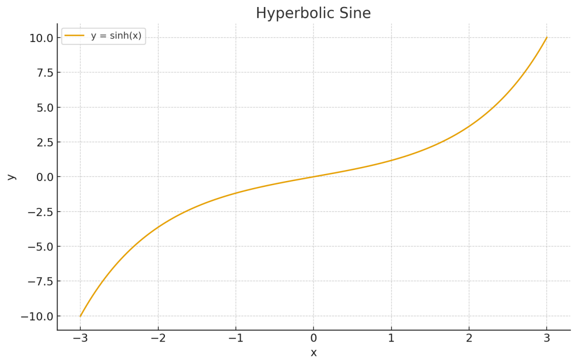
Basic properties of $\sinh x$

- Odd and strictly increasing on $\mathbb{R}$; range $(-\infty, \infty)$.
- Derivatives: $\dfrac{d}{dx}\sinh x = \cosh x \geq 1$ (with equality only at $x = 0$), $\dfrac{d^2}{dx^2}\sinh x = \sinh x$.
- Asymptotics: $\sinh x \sim \frac{1}{2}e^x$ as $x \to +\infty$ and $\sinh x \sim -\frac{1}{2}e^{-|x|}$ as $x \to -\infty$.
- Series (entire function):
$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}.$$
- Inverse and derivative:
$$\operatorname{arsinh} y = \ln\bigl(y + \sqrt{y^2+1}\,\bigr), \qquad \frac{d}{dy}\operatorname{arsinh} y = \frac{1}{\sqrt{1+y^2}}.$$
- Addition/angle-doubling:
$$\sinh(a \pm b) = \sinh a\, \cosh b \pm \cosh a\, \sinh b, \qquad \sinh(2x) = 2\sinh x\, \cosh x.$$
- Primitives: $\displaystyle\int \sinh x\, dx = \cosh x + C.$

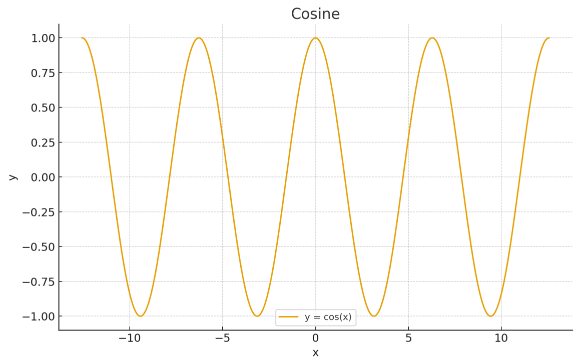
Basic properties of $\cos x$

- Even and 2π-periodic; range $[-1, 1]$.
- Zeros at $x = \frac{\pi}{2} + \pi k$; extrema: $\cos x = 1$ at $x = 2\pi k$, $\cos x = -1$ at $x = (2k+1)\pi$.
- Derivatives/ODE: $\dfrac{d}{dx}\cos x = -\sin x$, $\quad \dfrac{d^2}{dx^2}\cos x = -\cos x$ (solves $y'' + y = 0$).
- Series (entire function):
$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$
- Bounds and local shape (for small x): $1 - \frac{x^2}{2} \leq \cos x \leq 1$; more sharply, $1 - \frac{x^2}{2} \leq \cos x \leq 1 - \frac{x^2}{2} + \frac{x^4}{24}$.
- Addition/angle-doubling:
$$\cos(a \pm b) = \cos a\, \cos b \mp \sin a\, \sin b, \qquad \cos(2x) = 2\cos^2 x - 1 = 1 - 2\sin^2 x.$$
- Mean value over a period: $\dfrac{1}{2\pi}\displaystyle\int_0^{2\pi} \cos x\, dx = 0.$

Graphs of $\sinh x$ and $\cos x$



(a) $y = \sinh(x)$



(b) $y = \cos(x)$

Figure 8: Contrast: monotone unbounded hyperbolic vs. bounded periodic trigonometric.

Setup and notation

Let $z = x + iy \in \mathbb{C}$ and $w = re^{i\varphi} \in \mathbb{C} \setminus \{0\}$ be the polar form of w. Consider the holomorphic map

$$w = e^z = e^{x+iy} = e^x\, e^{iy}.$$

We write Arg for the principal argument (with a branch cut along the negative real axis) and log for the principal branch of the logarithm.

Coordinate relations: Cartesian $\leftrightarrow$ polar

From $w = e^x e^{iy}$ we immediately obtain

$$\boxed{r = |w| = e^x}, \qquad \boxed{\varphi = \arg w \equiv y \pmod{2\pi}}.$$

The inverse map (a multivalued logarithm) is

$$x = \ln r, \qquad y = \varphi + 2\pi k,\ \ k \in \mathbb{Z}, \qquad z = \ln r + i(\varphi + 2\pi k).$$

After restricting to a horizontal strip $y \in (\beta, \beta + 2\pi)$ we obtain a single-valued branch $z = \ln r + i\varphi$ for $\varphi \in (\beta, \beta + 2\pi)$.

Conformality, scale, and area element

Since $(e^z)' = e^z$, the map is conformal and the local scale factor is

$$\lambda(z) = |e^z| = e^x = r.$$

For infinitesimal displacements: $dr = r\, dx$ and $d\varphi = dy$. Line elements scale by λ and area elements by λ^2:

$$\mathrm{d}A_w = \lambda(z)^2\, \mathrm{d}A_z = |e^z|^2\, \mathrm{d}x\, \mathrm{d}y = r^2\, \mathrm{d}x\, \mathrm{d}y.$$

Level sets and orthogonality

Writing $w = e^z = e^x(\cos y + i \sin y) =: u(x,y) + iv(x,y)$, we have

$$u(x,y) = e^x \cos y, \qquad v(x,y) = e^x \sin y,$$

$$r = e^x, \qquad \varphi = y \pmod{2\pi}.$$

The level sets $x = \text{const}$ and $y = \text{const}$ (the rectangular grid in the z-plane) map to:

- $x = a \;\Rightarrow\; r = e^a$: concentric circles of radius e^a,
- $y = b \;\Rightarrow\; \varphi = b$: rays from the origin at angle b.

These two families are orthogonal both in the z- and w-planes (conformality preserves angles).

Images of lines and circles

Lines

- **Vertical:** $x = a$ map to circles $\{\,|w| = e^a\,\}$.
- **Horizontal:** $y = b$ map to rays $\{\,\arg w = b\,\}$.
- **Oblique:** $y = mx + b$ map to logarithmic spirals
$$w = e^x e^{i(mx+b)} \;\Rightarrow\; \arg w = b + m \ln|w| \quad \text{(log spiral)}.$$

Circles in z and their images

A general circle in the z-plane $z(\theta) = a + \rho e^{i\theta}$ yields

$$w(\theta) = e^a\, e^{\rho\cos\theta}\, e^{i(\Im a + \rho \sin\theta)}.$$

The radius $|w|$ oscillates between $e^{\Re a - \rho}$ and $e^{\Re a + \rho}$ and the angle varies as $\Im a + \rho \sin\theta$. In general this is *not* a circle (it is a smooth closed curve inside the annulus), except for the limiting case of a "circle through ∞" (i.e. a vertical line), which maps to a concentric circle in w.

Images of basic regions

- **Horizontal strips** $y \in (\beta, \beta + 2\pi)$: the image is $\mathbb{C} \setminus \{0\}$ minus a single ray at argument β (a slit plane).
- **Vertical strips** $x \in (a, b)$: the image is an *annulus*
$$A = \{\, e^a < |w| < e^b \,\}.$$
- **Rectangle** $R = \{a < x < b,\ \gamma < y < \delta\}$: the image is a *sector of an annulus*
$$\{\, e^a < |w| < e^b,\ \gamma < \arg w < \delta \,\}.$$
- **Half-plane** $x > a$: the image is the *exterior* of the disk $\{|w| > e^a\}$.
- **Disk** $\{|w| < R\}$: the preimage is the half-plane $x < \ln R$.

Inverse branch: the logarithm

The inverse of e^z is the multivalued logarithm:

$$\log w = \ln|w| + i(\arg w + 2\pi k), \quad k \in \mathbb{Z}.$$

Upon choosing a principal branch (e.g. cutting along $\arg w = \beta$), we get a single-valued holomorphic function Log_β on $\mathbb{C} \setminus \{re^{i\beta} : r \geq 0\}$ with

$$\mathrm{Log}_\beta(e^z) = z \quad \text{for } \Im z \in (\beta, \beta + 2\pi).$$

"Two grids" $\to$ a polar grid

The rectangular grid in z—the families $x = \text{const}$ and $y = \text{const}$—maps exactly to the polar grid in w: concentric circles $r = \text{const}$ and rays $\varphi = \text{const}$.

$$\{x = \text{const}\} \;\mapsto\; \{r = \text{const}\}, \qquad \{y = \text{const}\} \;\mapsto\; \{\varphi = \text{const}\}.$$

This is a textbook instance of conformality: angles between the families are preserved, and the scale grows like r as one moves away from 0.

Metric and harmonic features

- **Lengths:** a small segment of length ℓ in z maps to a segment of length $\approx r\,\ell$.
- **Areas:** a small area S maps to an area $\approx r^2 S$.
- **Cauchy–Riemann:** $u = e^x \cos y$, $v = e^x \sin y$ satisfy the C–R equations; u and v are harmonic and conjugate, hence their level sets are orthogonal.

Classical correspondences

1. **Rays** $\{\arg w = \varphi_0\} \leftrightarrow$ **horizontal lines** $\{y = \varphi_0 + 2\pi k\}$.
2. **Concentric circles** $\{|w| = R\} \leftrightarrow$ **vertical lines** $\{x = \ln R\}$.
3. **Annulus** $R_1 < |w| < R_2 \leftrightarrow$ **vertical strip** $\ln R_1 < x < \ln R_2$.
4. **Sector** $\{\alpha < \arg w < \beta\} \leftrightarrow$ **horizontal strip** $\{\alpha < y < \beta\}$.

Remark on the Riemann sphere

After compactification (adding the point ∞), a vertical line $x = a$ is a circle through ∞; its image under e^z is the circle $\{|w| = e^a\}$ in the w-plane (not passing through ∞). This explains why *only* those "circles through ∞" in z become circles in w.

Counterexample: a circle in z does not map to a circle in w

For $z(\theta) = a + \rho e^{i\theta}$ we have

$$|w(\theta)| = e^{\Re a + \rho \cos\theta}, \qquad \arg w(\theta) = \Im a + \rho \sin\theta,$$

which does not satisfy a circle equation in (u, v); the radius varies exponentially, producing a "rippled" closed curve within the annulus $e^{\Re a - \rho} \le |w| \le e^{\Re a + \rho}$.

Summary. The map $w = e^z$ is conformal, converting the rectangular grid to the polar grid and realizing the natural change $(x, y) \mapsto (r, \varphi) = (e^x, y)$. Vertical lines go to concentric circles, horizontal lines to rays, oblique lines to logarithmic spirals, and rectangles to annular sectors. Only "circles through ∞" in z (i.e. vertical lines) become circles in w; generic circles do not.